3 of 200 runs is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q1/5

6 + 18 runs out of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q2/5

True or false?

19 out of 200 is 19%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q3/5

34 of 200 is

A. 1.7%

B. 17%

C. 6.8%

D. 34%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q4/5

Is 2% less than 5%? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q5/5

SET 64 • HSS-IC.A.2

FORWARD 1

Design a simulation to test a model • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A player claims to make 70% of her free throws, and she made 11 of 20. Which simulation tests the claim?

A. Digits 0-6 = make; 20 digits per run; repeat many runs; find how often 11 or fewer makes occur.

B. Digits 0-6 = make; 20 digits in a single run; the claim is true if that run has 14 makes.

C. Digits 0-4 = make; 20 digits per run; repeat many runs; find how often exactly 11 makes occur.

D. Digits 0-6 = make; 11 digits per run; repeat many runs; find how often 20 makes occur.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q1/5

A player who claims to make 70% of free throws made 11 of 20. Why should a simulation count runs with 11 or fewer makes, not only runs with exactly 11?

A. Any one exact count can be rare; the question is how often results at least this far below the claim occur.

B. Runs with exactly 11 makes are impossible under the model, so they would never be counted at all.

C. The claim is about making 11 or fewer shots, not about making 70% of her free throws.

D. Counting 11 or fewer always makes the result significant, which is the goal of the simulation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q2/5

True or false?

A result that happens in 30% of simulated runs is strong evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q3/5

A common cutoff for "unusual" is

A. about 50% of runs

B. 100% of runs

C. about 5% of runs

D. about 25% of runs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q4/5

A simulation of a model is run 400 times instead of 200 times. What does this change?

A. It changes the observed data that the model is compared with.

B. It changes the probabilities that the model assigns to each outcome.

C. It turns an unusual result into a usual one, because more runs are counted.

D. It gives a more precise estimate of how often a result occurs under the model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q5

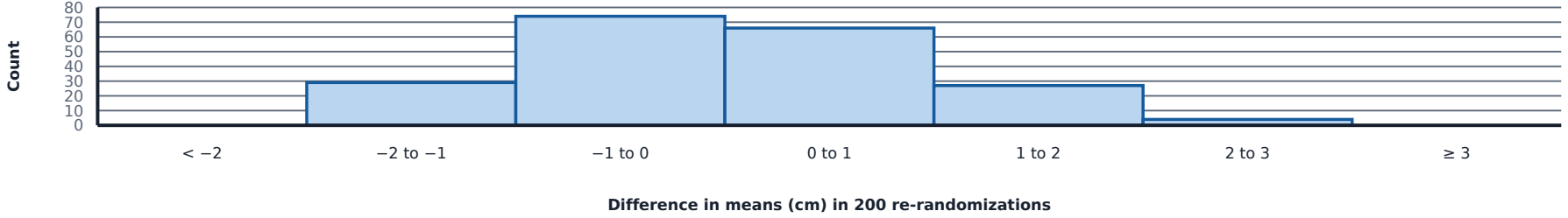
High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q5/5

SET 64 • HSS-IC.A.2

FORWARD 2

Compare two treatments with a re-randomization simulation • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In an experiment, 20 plants were randomly assigned to fertilizer or no fertilizer, and the fertilized plants grew 2.9 cm more on average. The histogram shows 200 re-randomized differences under the model that fertilizer has no effect; none was 2.9 cm or more. Use 5% as the cutoff for unusual. What do you conclude?



A. 0/200 < 5%: a difference this large is very unlikely by chance, so fertilizer appears to increase growth.

B. The difference is not significant, because no re-randomized difference matched it exactly.

C. 0/200: a difference this large has probability exactly 0, so the fertilizer is proved to work.

D. No conclusion is possible, because the result must first be compared with a fair coin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q1/5

What model does a re-randomization simulation assume?

A. The two groups were chosen by voluntary response.

B. The observed difference is exactly the true effect.

C. The treatment has no effect, so the group labels can be shuffled.

D. The treatment has the same large effect on every subject.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q2/5

True or false?

Re-randomization is a simulation of a model, like the coin-flip simulation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q3/5

Under the model of no treatment effect, 60 of 200 re-randomized differences were at least as large as the observed one. Use 5% as the cutoff for unusual. The result is

A. significant

B. proof of no effect

C. not statistically significant

D. impossible

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q4/5

How does a re-randomization test for an experiment extend the reasoning of testing a fair-coin claim with a simulation?

A. Both flip a real coin many times and then count how many heads appear in each of the groups of subjects in the study.

B. Both ask how often a model produces a result at least as extreme as the observed one; here the model is "no treatment effect."

C. The coin test uses a probability model, but the re-randomization test does not use any model of the results at all.

D. Both prove the model is true whenever the observed result turns out to be common among the simulated results.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q5/5

SET 65 • HSS-IC.B.3 • STATISTICS, INFERENCE, AND DECISIONS

Identify sample surveys, experiments, and observational studies

MAIN	Algebra II main question	**HSS-IC.B.3**
BACKWARD 1	Statistical questions and data collection	**6.SP.A.1**
BACKWARD 2	Association is not causation	**HSS-ID.C.9**
BACKWARD 3	Random samples	**7.SP.A.1**
FORWARD 1	Identify confounding and design flaws	**HSS-IC.B.3**
FORWARD 2	Analyze the results of an experiment	**HSS-IC.B.5**

SET 65 • HSS-IC.B.3

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Researchers randomly assign 100 volunteers to take a vitamin or a placebo and compare the number of colds. What type of study is this, and what is its purpose?

A. Experiment: to estimate the percent of all people who catch colds.

B. Observational study: to describe colds without any treatment.

C. Experiment: to test whether the vitamin causes fewer colds.

D. Sample survey: to estimate the percent of people who take vitamins.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q1/5

A pollster asks a random sample of 1,000 voters which candidate they prefer. What type of study is this, and what is its purpose?

A. Sample survey: to estimate the preference of all voters.

B. Experiment: to test whether a candidate causes votes.

C. Observational study: to show which candidate causes turnout.

D. Sample survey: to show the poll caused voters' choices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q2/5

Researchers compare the hospital records of smokers and nonsmokers for lung disease. What type of study is this, and can it by itself show causation?

A. Experiment; yes, smoking was the treatment in the study.

B. Observational study; no, the groups were not randomly assigned.

C. Observational study; yes, hospital records are accurate.

D. Sample survey; yes, the sample of records is large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q3/5

Which study best answers "Does playing music while studying improve test scores?"

A. a census of how many students in the school own headphones

B. an observational study of students who already choose to play music

C. a survey that asks students whether they think music helps them study

D. an experiment: randomly assign students to study with or without music and compare scores

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q4/5

Which study best answers "What percent of students at the school use the library weekly?"

A. a sample survey of a random sample of the school's students

B. a voluntary online poll posted on the library website

C. a randomized experiment that assigns students to use the library

D. an observational study of students in the library today

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 1

Statistical questions and data collection • Grade 6 • Statistics & Probability • 6.SP.A.1

Is "How many hours do students at our school sleep?" a statistical question?

A. No: it has a single correct answer.

B. Yes: the answers vary from student to student.

C. No: hours of sleep cannot be measured.

D. Yes: it asks about one particular student.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q1/5

Which method of collecting data best answers "What percent of the city's voters favor the new library?"

A. an experiment that assigns voters to read books

B. a survey of people leaving the old library

C. asking the members of the city council

D. a survey of a random sample of the city's voters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q2/5

True or false?

An experiment imposes a treatment on the subjects.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q3/5

Watching and recording without imposing treatments is

A. an experiment

B. an observational study

C. a census

D. a placebo

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q4/5

In an experiment, what is a placebo?

A. a random sample of the population

B. the group that gets the real treatment

C. a fake treatment with no active ingredient

D. a measurement taken before the study

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 2

Association is not causation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Ice cream sales and sunburns are positively correlated. Which explanation is best supported?

A. Sunburns cause ice cream sales, because a correlation shows the direction of cause.

B. The correlation must be a calculation error, because the variables are unrelated.

C. Ice cream causes sunburn, because the two variables are positively correlated.

D. Hot, sunny weather increases both; the correlation does not show that ice cream causes sunburn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q1/5

"Students with more books at home get higher test scores." Which is a plausible lurking variable?

A. family income

B. the test's page count

C. the student's ID number

D. the color of the book covers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q2/5

True or false?

A strong correlation proves that one variable causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q3/5

A variable that affects both variables in a study is

A. the placebo

B. the sample

C. a confounding (lurking) variable

D. the response

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q4/5

Which kind of study can show cause and effect?

A. a voluntary-response survey

B. an observational study

C. a randomized experiment

D. a sample survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 3

Random samples • Grade 7 • Statistics & Probability • 7.SP.A.1

Which method gives a simple random sample of 50 students from a school list?

A. Choose 50 students who are standing in the cafeteria line at lunch.

B. Number the students and use a random number generator to pick 50 different numbers.

C. Ask for 50 volunteers by posting a sign-up sheet in the school office.

D. Choose the first 50 names on the list, which is in alphabetical order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q1/5

Is surveying the students in the cafeteria line a random sample of the school?

A. Yes: the students in the line arrived in a random order.

B. No: a random sample must include every student.

C. No: students who are not in the line have no chance to be chosen.

D. Yes: any student could be in the cafeteria line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q2/5

True or false?

A random sample tends to represent the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q3/5

Which sample is random?

A. the teacher's friends among the students

B. students who volunteer when asked in class

C. names drawn from a hat with every student's name

D. the first 50 students to arrive at school

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q4/5

Why does a survey need a random sample?

A. so it needs to ask only 2 people

B. so every person gives the same answer

C. so it can show cause and effect by itself

D. so its results can be generalized to the population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q5/5

SET 65 • HSS-IC.B.3

FORWARD 1

Identify confounding and design flaws • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Students who choose to take an SAT prep course score higher than students who do not. Which is a confounding variable, with a correct reason?

A. Hair color: it differs between students, so it must confound the result of the study.

B. Motivation: students who choose the course may be more motivated, which also raises scores.

C. The test date: all students take the same test, so the date explains the difference.

D. The course itself: it raises scores, so it is the confounding variable in the study.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q1/5

Students who choose to take an SAT prep course score higher than students who do not. Which design could show whether the course causes higher scores?

A. Randomly assign volunteers to take the course or not, then compare their scores.

B. Let students choose whether to take the course again, but use a larger sample.

C. Compare the scores of students at two different schools that offer the course.

D. Survey many more students who chose to take the course about their scores.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q2/5

True or false?

An experiment without a control group can still show the treatment caused the result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q3/5

In an experiment, why use a placebo?

A. to make the study observational instead

B. to control for believing one is treated

C. to increase the sample size of the study

D. to make the real treatment work better

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q4/5

Why is an observational study, rather than an experiment, used to study whether smoking causes lung disease in people?

A. Experiments cannot use random assignment.

B. It would be unethical to assign people to smoke.

C. Smokers cannot be counted in an experiment.

D. Observational studies always show causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q5/5

SET 65 • HSS-IC.B.3

FORWARD 2

Analyze the results of an experiment • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

100 volunteers were randomly assigned to a vitamin or a placebo; the vitamin group had 1.2 fewer colds on average. Under the model that the vitamin has no effect, 40 of 200 re-randomizations gave the vitamin group at least 1.2 fewer colds. Use 5% as the cutoff for unusual. What do you conclude?

A. 40/200 = 20% > 5%: the difference could easily be chance, so it is not significant.

B. 40/200 = 20% > 5%: this proves the vitamin has no effect.

C. 40/200 = 20%: the vitamin prevents 20% of colds in these volunteers.

D. 40/200 = 20% > 5%: the difference is significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

In a randomized vitamin-versus-placebo experiment, the vitamin group had 1.2 fewer colds on average. Under the model that the vitamin has no effect, 2 of 200 re-randomized differences were that large in the vitamin's favor. Use 5% as the cutoff for unusual. What do you conclude?

A. 2/200 = 1%: this proves the vitamin works for everyone.

B. 2/200 = 1%: the vitamin works for only 1% of people.

C. 2/200 = 1% < 5%: significant, so the vitamin appears to reduce colds.

D. 2/200 = 1% < 5%: the difference is not significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

True or false?

Random assignment makes it possible to attribute a significant difference to the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

The re-randomization model assumes

A. the groups are different sizes

B. the treatment has no effect

C. the treatment works

D. the data are normal

Why can a significant difference in an observational study not be interpreted as a treatment effect the way it can in a randomized experiment?

A. Observational studies always use smaller samples, so their differences are always due to chance.

B. Observational studies cannot measure any differences between groups, so no difference can be found.

C. A significant difference in an observational study is always the result of a calculation error.

D. Without random assignment, the difference could come from confounding variables rather than the treatment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

SET 66 • HSS-IC.B.3 • STATISTICS, INFERENCE, AND DECISIONS

Explain random sampling versus random assignment

MAIN	Algebra II main question	**HSS-IC.B.3**
BACKWARD 1	Study types and their purposes	**HSS-IC.B.3**
BACKWARD 2	Choose a random sample	**7.SP.A.1**
BACKWARD 3	Correlation and causation	**HSS-ID.C.9**
FORWARD 1	Improve an experimental design	**HSS-IC.B.3**
FORWARD 2	Use re-randomization to judge a treatment difference	**HSS-IC.B.5**

SET 66 • HSS-IC.B.3

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

A random sample of 500 adults is surveyed about exercise and sleep; adults who exercise more tend to sleep better. Which conclusion is justified?

A. Exercise causes better sleep for all adults, because the sample was chosen at random.

B. The association can be generalized to all adults, but it does not show that exercise causes better sleep.

C. Nothing can be concluded about any adults beyond the 500 who were surveyed.

D. Exercise causes better sleep, but only for the 500 adults who were surveyed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q1/5

60 volunteers are randomly assigned to two diets, and diet A's group loses significantly more weight. Which conclusion is justified?

A. There is evidence that diet A caused more weight loss for these volunteers, but the result cannot be generalized to all people.

B. Diet A is associated with weight loss for all people, but the study cannot show that the diet caused it.

C. Diet A causes more weight loss for all people, because the volunteers were randomly assigned to the two diets.

D. Nothing can be concluded, because the volunteers chose to participate instead of being randomly sampled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q2/5

A random sample of 200 students from a school is randomly assigned to two study methods, and method 1 leads to significantly higher scores. Which conclusion is justified?

A. Method 1 is associated with higher scores, but the study cannot show that it causes them.

B. There is evidence that method 1 causes higher scores, but only for the 200 students in the study.

C. There is evidence that method 1 causes higher scores, and the result can be generalized to the school's students.

D. Method 1 causes higher scores for students at every school, because the result was significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q3/5

A teacher compares test scores of students in her class who chose to use a review app with those who did not. Which conclusion is justified?

A. The app causes higher scores, but only for the students in this class.

B. The app causes higher scores for all students, because the groups were compared.

C. The result can be generalized to all students, but it does not show causation.

D. Only an association in this class: there was no random assignment and no random sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q4/5

How can 40 subjects be randomly assigned to two groups of 20?

A. Put the first 20 subjects who arrive in group 1 and the other 20 subjects in group 2.

B. Put the 20 tallest subjects in group 1 and the 20 shortest subjects in group 2.

C. Let each subject choose a group, and stop when one of the groups has 20 subjects.

D. Number the subjects 1-40; choose 20 different random numbers for group 1; the rest form group 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 1

Study types and their purposes • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Which type of study uses random assignment of treatments?

A. an observational study

B. an experiment

C. a sample survey

D. a census

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q1/5

Which type of study is designed to estimate a population value?

A. a case study of one person

B. an experiment

C. a sample survey

D. a placebo

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q2/5

True or false?

An observational study assigns treatments.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q3/5

Random assignment is used to

A. avoid measuring

B. make the sample larger

C. make the groups comparable

D. choose who is in the population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q4/5

What does random sampling allow you to do?

A. generalize from the sample to the population

B. conclude cause and effect from the data

C. remove all chance variation from results

D. skip measuring the people in the sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 2

Choose a random sample • Grade 7 • Statistics & Probability • 7.SP.A.1

Which method chooses 20 names at random from a numbered list of 500?

A. Pick 20 different random numbers from 1 to 500 and take those names.

B. Pick 20 names that look like a good mix of the whole list.

C. Take the first 20 names on the numbered list, numbers 1 to 20.

D. Email all 500 and take the first 20 people who reply to the email.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q1/5

Is a sample of volunteers a random sample? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q2/5

True or false?

A sample of volunteers may differ from the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q3/5

Which method gives a random sample?

A. choosing the first 20 people on the list

B. drawing numbered slips from a well-mixed hat

C. asking for volunteers to sign up

D. asking friends to take part

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q4/5

Names are written on slips and drawn from a hat. Why must the slips be well mixed?

A. so the drawing takes longer

B. so the first slip drawn is always the same

C. so every slip has the same chance of being drawn

D. so the hat can hold more slips

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 3

Correlation and causation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Can an association between two variables in observational data prove causation? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q1/5

Students who exercise more tend to sleep better. Which is a plausible lurking variable?

A. the day of the month they were born

B. stress level

C. their locker number

D. the brand of their shoes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q2/5

True or false?

Random assignment controls lurking variables on average.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q3/5

Causation can be concluded from

A. an observational study

B. a correlation alone

C. a randomized experiment

D. a survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q4/5

Why does a correlation alone not show which variable causes the other?

A. Correlation can be found only for categories, not for numbers.

B. Correlation is always 0 when one variable causes the other.

C. Correlated variables can never be related to each other.

D. Either variable could affect the other, or a third variable could affect both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q5/5

SET 66 • HSS-IC.B.3

FORWARD 1

Improve an experimental design • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

60 volunteers try a new diet and lose weight. Why is a control group (for example a second, randomly assigned diet) needed to judge the diet?

A. Without a comparison group, the weight loss could be due to other factors, not the diet.

B. A control group makes every volunteer in the study lose more weight on the diet.

C. A control group proves that the new diet works for everyone who tries it.

D. A control group is needed only in studies with more than 100 volunteers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-F1-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q1/5

A drug study blocks subjects by age, then randomly assigns treatments within each age group. Why?

A. so the older subjects all get the drug and the younger the placebo

B. so every subject in each age group receives both treatments

C. so age differences cannot hide or create a treatment effect

D. so the study can avoid random assignment within the groups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q2/5

True or false?

In a double-blind experiment, neither subjects nor evaluators know who gets the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q3/5

60 volunteers were randomly assigned to two diets, and diet A led to more weight loss. Which change would let the result be generalized to all adults?

A. measuring the volunteers more often during the study

B. using a placebo for one group

C. adding more volunteers to each diet

D. randomly selecting subjects from all adults

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q4/5

Why are many subjects better than few in an experiment?

A. More subjects remove the need for random assignment to the groups.

B. More subjects always make the treatment itself work better.

C. More subjects make any observed result statistically significant.

D. Chance differences between groups are smaller, so real effects are easier to detect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q5/5

SET 66 • HSS-IC.B.3

FORWARD 2

Use re-randomization to judge a treatment difference • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

60 volunteers were randomly assigned to two diets; diet A's group lost 2.4 kg more on average. Under the model that the diets have the same effect, 5 of 200 re-randomizations gave a difference of at least 2.4 kg in favor of A. Use 5% as the cutoff for unusual. Is the difference significant?

A. No: 5/200 = 2.5% < 5%.

B. Yes: 5/200 = 2.5% < 5%.

C. No: 5/200 = 25% > 5%.

D. Yes: 5 is more than 2.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q1/5

60 volunteers (not a random sample of adults) were randomly assigned to two diets, and diet A's group lost significantly more weight. Which conclusion is justified?

A. Diet A is associated with weight loss, but the study cannot show that it caused it.

B. There is evidence that diet A caused more weight loss for these volunteers; it may not apply to all adults.

C. Nothing can be concluded, because the volunteers were not a random sample of adults.

D. Diet A causes more weight loss for all adults, because the result was significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q2/5

True or false?

Re-randomization shuffles the group labels as if the treatment had no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q3/5

If 70 of 200 re-randomized differences were at least 2.4 kg, conclude

A. the difference could be due to chance

B. the study was observational

C. diet B works

D. diet A works

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q4/5

Why is a re-randomization test valid only when the original groups were formed by random assignment?

A. Without random assignment the group means cannot be computed, so there is nothing to shuffle in the test.

B. Random assignment makes every difference significant, so the test can only be used after random assignment.

C. Re-randomization needs exactly the same number of subjects as a fair coin needs flips, so it requires random groups.

D. The shuffle copies the random assignment; if the groups were not randomly formed, the shuffle does not model how they arose.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q5/5

SET 67 • HSS-IC.B.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate a population proportion with a margin of error from a simulation

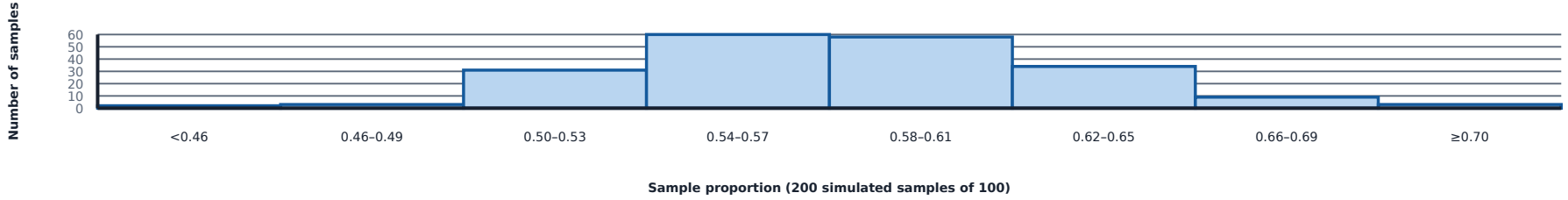
MAIN	Algebra II main question	**HSS-IC.B.4**
BACKWARD 1	Statistics, parameters, and sampling variability	**HSS-IC.A.1**
BACKWARD 2	Proportions and percents	**6.RP.A.3.c**
BACKWARD 3	Standard deviation as typical distance from the mean	**HSS-ID.A.2**
FORWARD 1	Plan a sample size for a desired margin of error	**HSS-IC.B.4**
FORWARD 2	Evaluate reports that use margins of error	**HSS-IC.B.6**

SET 67 • HSS-IC.B.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

In a random sample of 100 students, 58 want a later start time. The histogram shows 200 simulated samples of 100 from a population with p = 0.58; their standard deviation is 0.049. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval estimate, as decimals.



Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q1/5

In a random sample of 300 voters, 120 support a measure. Simulated sample proportions (n = 300, p = 0.40) have standard deviation 0.0275. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the estimate, the margin of error, and the interval, as decimals.

Estimate: _____

Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q2/5

In a random sample of 50 shoppers, 35 use a store app. Simulated sample proportions (n = 50, p = 0.70) have standard deviation 0.064. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval, as decimals.

Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q3/5

Three random-sample surveys have simulated-proportion standard deviations 0.064 (n = 50), 0.049 (n = 100), and 0.0275 (n = 300). Using margin of error = 2 SD, which survey has the smallest margin of error, and why?

A. n = 100 (margin 0.098): the middle size balances the two kinds of error.

B. All are equal: the margin depends only on the population size.

C. n = 300 (margin 0.055): larger samples give less variable sample proportions.

D. n = 50 (margin 0.128): smaller samples are easier to measure exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q4/5

A report from a random sample says "52% of voters support the plan, with a margin of error of 3 percentage points (95% confidence)." Which interpretation is correct?

A. The percent of all voters who support the plan is plausibly between 49% and 55%.

B. Exactly 52% of all voters support the plan, give or take nothing at all.

C. The plan will win, because 52% is more than 50% of the voters.

D. Between 49% and 55% of the sampled voters support the plan.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q5/5

SET 67 • HSS-IC.B.4

BACKWARD 1

Statistics, parameters, and sampling variability • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

In a random sample of 100 students, 58 want a later start time. What is the statistic?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q1/5

In a random sample of 100 students, 58% want a later start time. What parameter does 58% estimate?

A. the margin of error of the survey

B. the percent of all students who want a later start

C. the number of students in the sample

D. the percent of the 100 sampled students who want it

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B1-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q2/5

True or false?

Different random samples of 100 would give different sample proportions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q3/5

A histogram of 200 simulated sample proportions (samples of 100) shows

A. the population parameter exactly

B. how sample proportions vary

C. the whole population of values

D. one sample of 100 values

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q4/5

Why must a sample be random for its margin of error to be meaningful?

A. A random sample always has a margin of error of 0, so any error is bias.

B. A biased sample has a smaller margin of error than a random one.

C. The margin describes chance variation in random samples; a biased sample can be off by more.

D. The margin depends only on the size of the population, not the sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • HSS-IC.B.4

BACKWARD 2

Proportions and percents • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write 120 out of 300 as a proportion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q1/5

Write 0.58 ± 0.10 as an interval.

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q2/5

True or false?

35 out of 50 is 35%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q3/5

52% ± 3% gives

A. 52% to 55%

B. 49% to 55%

C. 50% to 54%

D. 49% to 52%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q4/5

Write 0.055 as a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSS-IC.B.4

BACKWARD 3

Standard deviation as typical distance from the mean • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

By the 68-95-99.7 rule, what percent of a normal distribution lies within 2 standard deviations of the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q1/5

If the SD is 0.049, what is 2 SDs?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q2/5

True or false?

A smaller standard deviation means more variability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q3/5

Which set of simulated proportions varies least?

A. SD 0.1

B. SD 0.0275

C. SD 0.049

D. SD 0.064

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q4/5

Compute 2×0.064.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q5/5

SET 67 • HSS-IC.B.4

FORWARD 1

Plan a sample size for a desired margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A survey of a random sample of 100 students has a margin of error of 0.10. Quadrupling the sample size halves the margin of error. What is the margin of error with n = 400? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q1/5

About what sample size gives a margin of error of 0.05 when $p \approx 0.5$? (Use margin $\approx 2\sqrt{(p(1-p)/n)}$.)

$n \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q2/5

True or false?

Doubling the sample size halves the margin of error.

A. True

B. False

A random sample of n = 100 people gives a margin of error for a proportion. Which change to the sample size reduces the margin of error the most?

A. use n = 900

B. use n = 200

C. use n = 110

D. take a second sample of 100 and report it alone

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F1-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q4/5

Two random-sample polls of the same population have margins of error 0.05 and 0.025 for a proportion. The margin of error is about proportional to $1/\sqrt{n}$. How many times as large is the second sample as the first?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSS-IC.B.4

FORWARD 2

Evaluate reports that use margins of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A poll shows candidate A at 51% and B at 49%, each with a margin of error of ±4 percentage points. A headline says "A leads." Evaluate.

A. Justified: the margin of error is the same for both, so it cancels and A leads by 2.

B. Not justified: A must have less than 50% support, since 51% − 4% is below 50%.

C. Justified: 51% is greater than 49%, and the poll used a large sample.

D. Not justified: the intervals 47%−55% and 45%−53% overlap, so either could lead.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

A report gives 58% ± 10% from a random sample of 100 students and claims "a majority of students want a later start." Evaluate.

A. Not clearly justified: the interval 48% to 68% includes values below 50%.

B. Justified: 58% is more than 50%.

C. Justified: the whole interval is above 50%.

D. Not justified: a sample of 100 can never support a claim.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

True or false?

A margin of error accounts for bias from a voluntary-response sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

Which claim is best supported by its data?

A. A random sample gives 52% ± 4% support, so exactly 52% support it.

B. A random sample gives 70% ± 3% support, so a majority support the plan.

C. A random sample gives 51% ± 5% support, so a majority support the plan.

D. An online poll gives 60% support, so a majority support the plan.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

Besides the percent, what should a careful poll report include?

A. the sample size, how the sample was chosen, and the margin of error

B. only the percent, rounded to the nearest whole number

C. only last year's result, so readers can compare

D. the names and answers of the people who were surveyed

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

SET 68 • HSS-IC.B.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate a population mean with a margin of error from a simulation

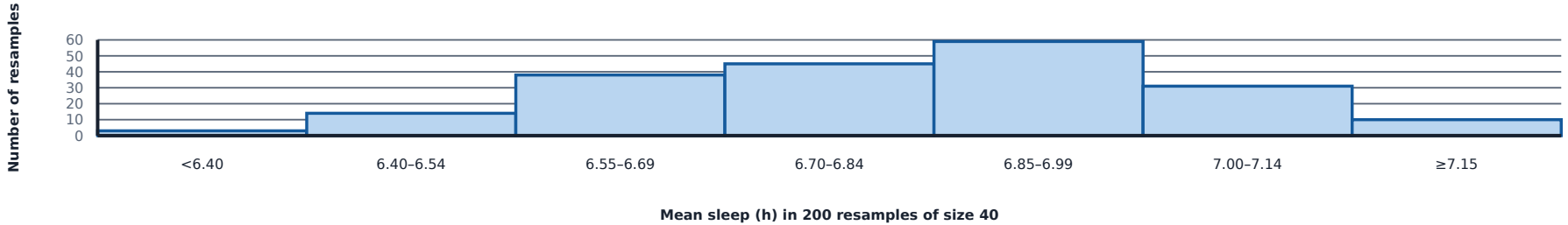
MAIN	Algebra II main question	**HSS-IC.B.4**
BACKWARD 1	Mean of a data set	**6.SP.B.5.c**
BACKWARD 2	What resampling simulates	**HSS-IC.A.1**
BACKWARD 3	Two standard deviations and the 95% range	**HSS-ID.A.2**
FORWARD 1	Use intervals to judge claims about a mean	**HSS-IC.B.4**
FORWARD 2	Compare two group means with re-randomization	**HSS-IC.B.5**

SET 68 • HSS-IC.B.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample of 40 students slept a mean of 6.8 hours. The histogram shows the means of 200 resamples of the data; their standard deviation is 0.20 hours. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval, in hours.



Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q1/5

A random sample of 25 apples has mean weight 152 g. Simulated sample means have standard deviation 3.6 g. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval, in grams.

Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q2/5

A random sample of 60 commuters has mean commute 24.5 minutes. Simulated sample means have standard deviation 1.4 minutes. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval, in minutes.

Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q3/5

A random sample of 40 students gives the 95% interval 6.4 to 7.2 hours for the mean hours of sleep. Which interpretation is correct?

A. Every student at the school sleeps at least 6.4 hours each night.

B. The 40 sampled students each slept between 6.4 and 7.2 hours.

C. The mean sleep of all students is plausibly between 6.4 and 7.2 hours.

D. 95% of all students sleep between 6.4 and 7.2 hours each night.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q4/5

A random sample of 25 apples gives a margin of error of 7.2 g for the mean weight. Assume apple weights vary the same way and the same 2-SD (95%) convention is used, so the margin of error is proportional to $1/\sqrt{n}$. What margin of error do you expect from a random sample of 100 apples? Give grams.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 1

Mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of 6, 7, 7.5, 6.5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q1/5

What does the mean of a random sample estimate?

A. the margin of error

B. the population mean

C. the largest value in the population

D. the sample size

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q2/5

True or false?

The mean is not affected by extreme values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q3/5

The mean of 150, 152, 154 is

A. 154

B. 150

C. 152

D. 456

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q4/5

Each of 40 values in a data set increases by 0.5. By how much does the mean increase? Give a number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 2

What resampling simulates • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

Why can resampling a random sample (drawing with replacement) show how sample means vary?

A. Resampling collects new data from the whole population each time it is done.

B. Resampling always gives the true population mean as the center.

C. Resampling removes all chance variation from the sample means.

D. The resamples act like new samples from a population that looks like the sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q1/5

In a histogram of 200 resample means, what does one bar's height show?

A. the chance the population mean is in that interval

B. the mean of the original sample

C. how many of the 200 resample means fell in that interval

D. the number of values in each resample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q2/5

True or false?

Sample means vary less than individual values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q3/5

A random sample of 40 students slept a mean of 6.8 hours. The parameter being estimated is

A. 6.8

B. the mean sleep of all students

C. the 40 students' mean

D. the resample SD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q4/5

A random sample of 40 students had a mean of 6.8 hours of sleep. Is 6.8 hours a parameter or a statistic?

A. a statistic: it describes the sample of 40

B. a parameter: it describes all students

C. a statistic: it describes all students

D. a parameter: it describes the sample of 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 3

Two standard deviations and the 95% range • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Compute $6.8 - 2(0.20)$ and $6.8 + 2(0.20)$.

$6.8 - 2(0.20) =$ _____

$6.8 + 2(0.20) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q1/5

Compute 2×1.4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q2/5

True or false?

About 95% of a mound-shaped distribution is within 2 SDs of its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q3/5

152 ± 7.2 gives

A. 144.8 to 159.2

B. 145 to 159

C. 152 to 159.2

D. 144.8 to 152

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q4/5

Is 7.5 inside the interval from 6.4 to 7.2? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q5/5

SET 68 • HSS-IC.B.4

FORWARD 1

Use intervals to judge claims about a mean • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample gives the 95% interval 6.4 to 7.2 hours for students' mean sleep. A health guide says students average 8 hours. Does the interval support this?

A. Yes: some of the students in the sample sleep 8 hours or more.

B. No: 8 is outside the interval of plausible values for the mean.

C. Yes: 8 is close to the upper end of the interval, 7.2 hours.

D. No: the interval proves the mean is exactly its center, 6.8 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q1/5

A random sample gives the 95% interval 144.8 g to 159.2 g for the mean apple weight. A grower claims the apples average 155 g. Is the claim consistent with the interval?

A. No: the mean must be exactly 152 g.

B. Yes: the claim is proved true because 155 is inside it.

C. No: 155 g is not the center of the interval.

D. Yes: 155 g is inside the interval, so it is a plausible mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q2/5

True or false?

A value inside a 95% interval estimate for a mean is proved to be the true mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q3/5

A random sample of 60 commuters gives an interval for the mean commute time. Which change would most likely narrow the interval?

A. sampling 240 commuters instead of 60

B. using the median

C. rounding the data

D. sampling 30 commuters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q4/5

A random sample of students at one school gives the interval 6.4 to 7.2 hours for their mean sleep. Why should the interval not be applied to students at a different school?

A. The students at the other school must have a higher mean sleep time.

B. An interval for a mean can only describe a single student's sleep.

C. The interval is too narrow to be applied to any group of students.

D. The sample came only from this school, so it estimates only this school's mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSS-IC.B.4

FORWARD 2

Compare two group means with re-randomization • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In an experiment, 40 students were randomly assigned to use a study app or not (20 each). The app group scored 6 points more on average. Under the model that the app has no effect, 9 of 200 re-randomizations gave the app group at least 6 more points. Using a one-sided count and a 5% cutoff, is the difference significant?

A. No: 9/200 = 4.5% > 5%.

B. Yes: 9/200 = 4.5% < 5%.

C. No: 9 is more than 6.

D. Yes: 9/200 = 45%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q1/5

Students were randomly assigned to a treatment group (study app) or a control group (no app), and the difference in mean scores was re-randomized 200 times. What null model do the re-randomizations represent?

A. The app has no effect on scores, so the group labels are interchangeable.

B. The app raises every score by exactly 6 points.

C. The students were chosen by voluntary response.

D. The control group scores lower in every re-randomization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q2/5

True or false?

An interval for one mean and a re-randomization test for two means both use simulated variability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q3/5

In an app experiment, the app group scored 6 points more on average. Under the model of no app effect, 30 of 200 re-randomizations gave the app group at least 6 more points. Use 5% as the cutoff for unusual. The conclusion is

A. the difference could be chance (15% > 5%)

B. the app works (15% < 50%)

C. the app is proved to have no effect

D. the app lowers scores

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q4/5

How is the margin of error for one mean related to judging a difference between two means?

A. A difference between two means never varies by chance, so it does not need any measure of chance variation.

B. The margin of error for one mean is always equal to the difference between the two group means.

C. Both ask how much a statistic varies by chance; a difference is meaningful only if it is large compared with that variability.

D. They are unrelated: one uses the data, and the other uses only the number of subjects in each group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q5/5

SET 69 • HSS-IC.B.5 • STATISTICS, INFERENCE, AND DECISIONS

Compare two treatments with a re-randomization simulation

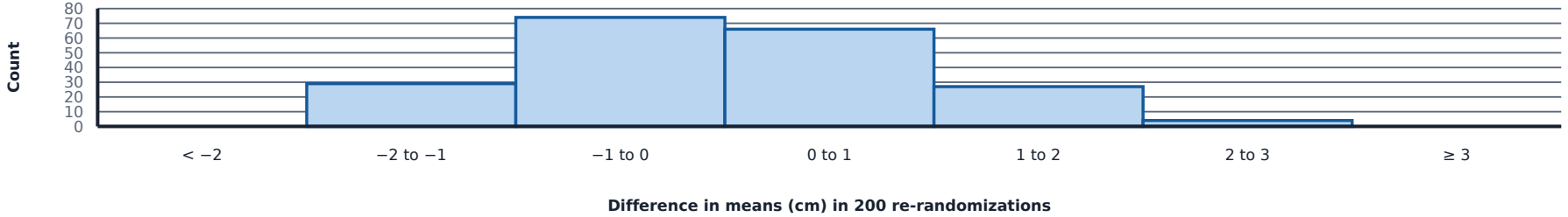
MAIN	Algebra II main question	**HSS-IC.B.5**
BACKWARD 1	Compare the means of two groups	**7.SP.B.4**
BACKWARD 2	Random assignment in experiments	**HSS-IC.B.3**
BACKWARD 3	Probability as relative frequency in simulations	**7.SP.C.6**
FORWARD 1	Two-sided comparisons and design choices	**HSS-IC.B.5**
FORWARD 2	Evaluate reports of experiments	**HSS-IC.B.6**

SET 69 • HSS-IC.B.5

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

10 plants got fertilizer and 10 did not (randomly assigned). Fertilized plants grew 2.9 cm more on average. The histogram shows 200 re-randomized differences under the model that fertilizer has no effect. Use a one-sided count (2.9 cm or more) and a 5% cutoff. Which conclusion is justified?



A. Most re-randomized differences were near 0, so the observed difference is not significant.

B. 4 of 200 (2%) were 2 cm or more, so the observed difference of 2.9 cm is not significant.

C. 0 of 200 were 2.9 cm or more; 0% < 5%, so the difference is significant: fertilizer appears to increase growth.

D. 0 of 200 were 2.9 cm or more, so this difference has probability exactly 0 and fertilizer is proved to work.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q1/5

In a randomized caffeine experiment, the caffeine group's mean reaction time was 0.04 s faster. Under the model that caffeine has no effect, 62 of 200 re-randomizations gave the caffeine group a mean at least 0.04 s faster. Using a one-sided count and a 5% cutoff, is the difference significant?

A. No: 62/200 = 31% > 5%, so the difference could easily be chance.

B. Yes: 62/200 = 31% is more than 5%, so the difference is significant.

C. No: 31% > 5%, and this proves that caffeine has no effect.

D. Yes: 0.04 s is less than 5% of a second, so it is significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q2/5

Students were randomly assigned to use a study app or not, and the app group scored 6 points higher on average. Under the model that the app has no effect, 9 of 200 re-randomizations gave the app group at least 6 more points. Using a one-sided count and a 5% cutoff, is the difference significant?

A. Yes: 9/200 = 45%, which is above 5%.

B. No: 9 is more than 6, so the app has no effect.

C. No: 9/200 = 4.5%, which is above 5%.

D. Yes: 9/200 = 4.5%, which is below 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q3/5

What does a re-randomization simulation assume, and how is it carried out?

A. It assumes no treatment effect; each group's values are re-sorted from smallest to largest, and the two sorted lists are compared once to find the difference.

B. It assumes the treatment works; the observed difference is doubled and halved many times, and the results are graphed to show how large the effect could be.

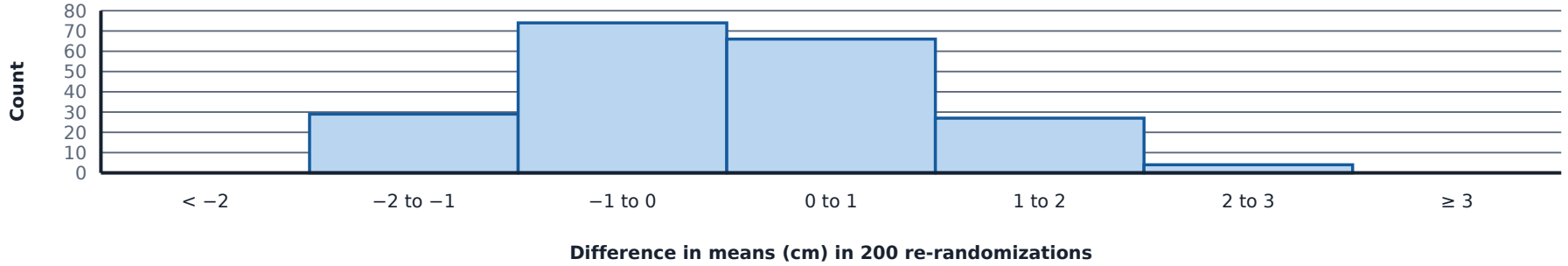
C. It assumes no treatment effect; all responses are pooled and randomly re-split into groups of the original sizes, and the difference in means is recorded many times.

D. It assumes the treatment works; new subjects are recruited for each run, and the difference in their group means is recorded many times to estimate the true effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q4/5

Using the histogram of 200 re-randomized differences, how many differences were 2 cm or more, and would an observed difference of 2 cm be significant using a one-sided count and a 5% cutoff?



A. 0 of 200; a difference of 2 cm has probability 0 under the model.

B. 4 of 200 (2%); 2% < 5%, so yes, it would be significant.

C. 31 of 200 (15.5%); 15.5% > 5%, so no, it would not be significant.

D. 4 of 200 (2%); 2% < 5%, so no, it would not be significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 1

Compare the means of two groups • Grade 7 • Statistics & Probability • 7.SP.B.4

Group A has mean 25.4 cm and group B has mean 22.5 cm. Find the difference A − B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q1/5

Group A: 4, 6, 8. Group B: 3, 5, 4. Find the difference in means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q2/5

True or false?

A difference in sample means could happen even if the treatments have the same effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q3/5

Mean A = 0.52 s and mean B = 0.48 s. A − B is

A. −0.04 s

B. 0.4 s

C. 1.00 s

D. 0.04 s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q4/5

Why is a difference between two group means compared with the variability in the groups?

A. Any difference between the group means proves a treatment effect.

B. The variability in the groups gives the exact size of the effect.

C. A difference is meaningful only if it is large relative to the natural variation.

D. Variability matters only when the two group means are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 2

Random assignment in experiments • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

In a fertilizer experiment, 20 plants were randomly assigned to fertilizer or no fertilizer. Why were they randomly assigned?

A. to make the groups comparable, so a difference can be due to the fertilizer

B. so that no measurements of growth are needed afterward

C. so the result can be generalized to all plants of every kind, everywhere

D. to make sure every plant grows exactly the same amount

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-B2-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q1/5

How can 20 plants be randomly assigned to two groups of 10?

A. Put the 10 tallest plants in the fertilizer group.

B. Let the gardener pick 10 healthy-looking plants for fertilizer.

C. Put the 10 plants nearest the window in the fertilizer group.

D. Number the plants 1-20 and use random numbers to choose 10 for fertilizer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-B2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q2/5

True or false?

Without random assignment, a re-randomization test does not justify a cause-and-effect conclusion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q3/5

Random assignment is a feature of

A. censuses

B. experiments

C. observational studies

D. surveys

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q4/5

A randomized experiment gives a significant difference. Which conclusion does it allow?

A. the difference is due to chance alone, not the treatment

B. there is evidence the treatment caused the difference for these subjects

C. the treatment causes the same difference in everyone everywhere

D. the subjects were a random sample of everyone in the population

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 3

Probability as relative frequency in simulations • Grade 7 • Statistics & Probability • 7.SP.C.6

62 of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q1/5

9 of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q2/5

True or false?

An event that happens in 40% of simulated runs is unusual.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q3/5

0 of 200 runs suggests the observed result is

A. very unlikely under the model

B. certain

C. impossible

D. typical

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q4/5

Is 4/200 less than 5%? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q5/5

SET 69 • HSS-IC.B.5

FORWARD 1

Two-sided comparisons and design choices • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In a caffeine experiment, the observed difference in mean reaction time was 0.04 s. The question is whether caffeine changes reaction time in either direction. Which re-randomized differences should be counted?

A. only those with difference ≥ 0.04 s

B. those with |difference| ≥ 0.04 s, in either direction

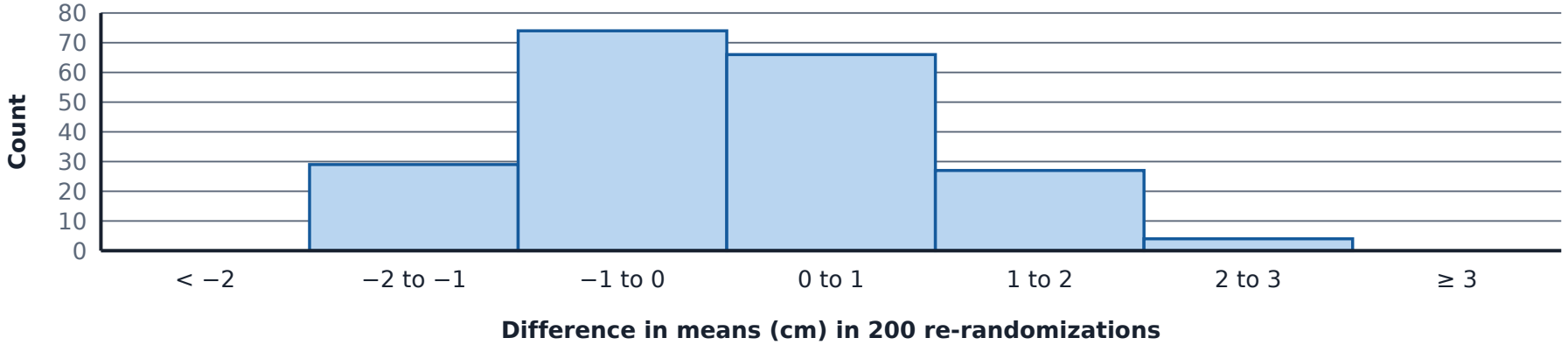
C. only those with |difference| ≤ 0.04 s

D. only those with difference exactly 0.04 s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F1-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q1/5

The histogram shows 200 re-randomized differences in mean growth (cm). How many differences were 2 cm or more, or less than −2 cm?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q2/5

True or false?

Using more plants in each group usually makes the re-randomized differences less spread out.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q3/5

A small observed difference can be significant when

A. the groups have very little natural variation

B. the data are rounded to whole numbers

C. the groups were not randomly assigned

D. the sample in each group is small

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q4/5

Why should the cutoff for significance (such as 5%) be chosen before seeing the data?

A. The cutoff changes the data that are collected in the study.

B. The cutoff must equal the observed proportion of the data.

C. Choosing it afterward could bias the conclusion toward the result one wants.

D. Cutoffs chosen in advance are always 50%, the fairest choice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q5/5

SET 69 • HSS-IC.B.5

FORWARD 2

Evaluate reports of experiments • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A company says its supplement "improved memory scores by 3 points" in a study of 8 people with no control group. Which evaluation is correct?

A. Strong evidence: an increase of 3 points is a real improvement in memory for these people.

B. Weak evidence, but only because an increase of 3 points is too small a number to matter.

C. Strong evidence: every one of the 8 people in the study took the supplement as directed.

D. Weak evidence: with no control group and only 8 people, the change could be due to other factors or chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q1/5

A report says a new teaching method "significantly" raised scores in a randomized experiment. Under the no-effect model, 2 of 200 re-randomized differences were at least as large as the observed difference. Use 5% as the cutoff for unusual. Evaluate the report.

A. Supported: the assignment was random, and 2/200 = 1% < 5%.

B. Supported: the method is proved to work for all students.

C. Not supported: re-randomization cannot be used for experiments.

D. Not supported: 2/200 = 1% is too small to matter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q2/5

True or false?

"Statistically significant" means the effect is large and important.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q3/5

Which report is most convincing?

A. a study of 500 users with no comparison group or random assignment

B. testimonials from many satisfied customers

C. a randomized experiment with a control group and a rare result

D. an online poll of 10,000 users of the product

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q4/5

Which information should a report of an experiment include?

A. only the observed difference between the two groups

B. the assignment method, group sizes, observed difference, and how unusual it is

C. only the conclusion that the treatment works for everyone

D. only the names of the treatments, the date of the study, and who paid for it

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q5/5

SET 70 • HSS-IC.B.6 • STATISTICS, INFERENCE, AND DECISIONS

Evaluate claims in reports based on data

MAIN	Algebra II main question	**HSS-IC.B.6**
BACKWARD 1	Random samples and bias	**HSS-IC.A.1**
BACKWARD 2	Study design and causation	**HSS-IC.B.3**
BACKWARD 3	Read scales on graphs	**3.MD.B.3**
FORWARD 1	Rewrite claims so that they follow from the data	**HSS-IC.B.6**
FORWARD 2	Analyze decisions based on test results	**HSS-MD.B.7**

SET 70 • HSS-IC.B.6

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

An ad says "4 out of 5 dentists recommend our toothpaste." Only 5 dentists, chosen by the company, were asked. Evaluate the claim.

A. Weak: the sample is tiny and not random, so it cannot represent all dentists.

B. Strong: dentists are experts, so any group of them is representative.

C. Weak: the claim would only be fine if all 5 of the dentists agreed.

D. Strong: 80% is a large majority of the dentists who were asked.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q1/5

A report says "Students who eat breakfast earn higher grades, so breakfast causes higher grades." The data came from a survey. Evaluate.

A. Justified: the survey found that students who eat breakfast have higher grades.

B. Not justified: breakfast and grades cannot be associated in a survey.

C. Not justified: a survey shows association, not causation; lurking variables could explain it.

D. Justified: the survey was large enough to show cause and effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q2/5

A bar graph shows sales of 50 and 52 thousand dollars, with the vertical axis starting at 48 thousand, so one bar looks twice as tall. Evaluate the impression the graph gives.

A. Fair: starting the axis at 48 thousand makes the graph easier to read.

B. Fair: the tops of the bars show the correct values of 50 and 52 thousand.

C. Misleading: the taller bar should be drawn more than twice as tall.

D. Misleading: the truncated axis makes a 4% difference look like a 100% difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q3/5

A TV show asks viewers to text their vote; 90% of 10,000 texts favor a policy. The show claims most citizens support it. Evaluate.

A. Justified: 90% is so far above 50% that the method of collecting the votes does not matter.

B. Not justified: texters are a voluntary response, not a random sample.

C. Justified: 10,000 texts form a very large sample of the citizens.

D. Not justified: 10,000 texts is too few to represent a whole country.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q4/5

A poll shows candidate A at 51% and B at 49%, each with a margin of error of ±4 percentage points. A headline says "A is winning." Evaluate.

A. Not justified: the intervals 47%−55% and 45%−53% overlap, so the lead is within the margin of error.

B. Not justified: B is actually winning, because the margin favors the trailing candidate.

C. Justified: the margin of error is the same for both candidates, so it cancels out.

D. Justified: A has more support than B in the poll, so A is winning the race.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 1

Random samples and bias • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

Why is a voluntary-response sample usually biased?

A. Every member of the population responds to it.

B. Voluntary samples are always too small to use.

C. The sample is chosen with random numbers instead.

D. People with strong opinions are more likely to respond.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q1/5

What does a random sample allow?

A. concluding cause and effect from the data

B. skipping the margin of error in the report

C. generalizing from the sample to the population

D. eliminating all chance variation from results

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q2/5

True or false?

A very large sample cannot be biased.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q3/5

Which sample is most likely biased?

A. customers who chose to write reviews

B. a random sample of 1,000

C. randomly chosen customers

D. every customer

Is "5 dentists chosen by the company" a random sample? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 2

Study design and causation • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

A survey records students' breakfast habits and grades. What type of study is it?

A. an observational study

B. a randomized experiment

C. a placebo-controlled experiment

D. a simulation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q1/5

Which design could test whether eating breakfast causes higher grades?

A. a voluntary online poll about eating breakfast and grades

B. a larger survey of students' breakfast habits and grades

C. a randomized experiment that assigns students to eat breakfast or not

D. comparing grades of students who already eat breakfast or not

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q2/5

True or false?

Only randomized experiments can establish cause and effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q3/5

A confounding variable is

A. the response variable that is measured in the study

B. a variable related to both the explanatory and response variables

C. the number of subjects in the sample of the study

D. the margin of error of the estimate

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q4/5

Students who eat breakfast tend to earn higher grades. Which is a plausible confounding variable?

A. the students' locker numbers

B. the amount of sleep students get

C. the color of the cereal box

D. the day the survey was printed

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 3

Read scales on graphs • Grade 3 • Measurement & Data • 3.MD.B.3

A bar graph's vertical axis starts at 48. One bar reaches 50 and another reaches 52. What values do the two bars show?

First bar: _____

Second bar: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q1/5

What percent larger is 52 than 50?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q2/5

True or false?

An axis that does not start at 0 can make differences look larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q3/5

On a graph whose axis starts at 48, a bar of 52 looks how many times as tall as a bar of 50?

A. the same

B. 1.04 times

C. 2 times

D. 4 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q4/5

A bar graph of sales of 50 and 52 thousand dollars has a vertical axis starting at 48 thousand. How should it be redrawn to be fair?

A. Make the bars wider.

B. Remove the axis labels.

C. Start the axis at 51 to show the difference better.

D. Start the vertical axis at 0, or clearly mark the axis break.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q5/5

SET 70 • HSS-IC.B.6

FORWARD 1

Rewrite claims so that they follow from the data • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A survey found that students who eat breakfast earn higher grades. Which claim is supported by the survey?

A. Higher grades cause students to eat breakfast.

B. Eating breakfast causes higher grades.

C. Students who eat breakfast tend to have higher grades.

D. Skipping breakfast causes lower grades.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q1/5

A poll shows 51% for A and 49% for B, each with a margin of error of ±4 percentage points. Which headline is accurate?

A. "The race is too close to call"

B. "B is winning"

C. "A is winning"

D. "A has a certain 2-point lead"

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q2/5

True or false?

Saying "is associated with" instead of "causes" is appropriate for an observational study.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q3/5

Which conclusion is justified by a random sample of 1,000 voters showing 62% ± 3% support?

A. the sample was biased

B. most voters support the measure

C. exactly 62% support it

D. the measure will pass by 24 points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q4/5

An ad says "4 out of 5 dentists recommend our toothpaste," based on 5 dentists chosen by the company. What would make the claim more convincing?

A. printing the claim in larger letters on the box

B. a large random sample of dentists, with the question wording reported

C. asking the same 5 dentists again a year later

D. choosing 5 different dentists who already like the brand

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q5/5

SET 70 • HSS-IC.B.6

FORWARD 2

Analyze decisions based on test results • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A screening test detects 95% of people who have a disease (sensitivity 95%) and gives a false positive for 5% of healthy people. In a group of 10,000 people, 1% have the disease. How many true positives are there, and how many positive results in all?

True positives: _____

All positives: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q1/5

A screening test has sensitivity 95% and a 5% false-positive rate. In a group of 10,000 people where 1% have the disease, there are 95 true positives and 495 false positives. A report says "a positive test means you almost surely have the disease." Evaluate.

A. Correct: only 5% of positive results are false.

B. Misleading: 495/590 ≈ 84% of positives have the disease.

C. Correct: the test detects 95% of cases.

D. Misleading: only 95/590 ≈ 16% of positives have the disease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q2**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q2/5

True or false?

The probability of having the disease given a positive result can be low even with an accurate test.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q3/5

A screening test has sensitivity 95% and a 5% false-positive rate. If 1% of a population has the disease, P(disease | positive) is about

A. 0.99

B. 0.05

C. 0.95

D. 0.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q4/5

For a screening test, P(disease | positive) ≈ 0.16. Which follow-up decision makes the most sense after a positive result?

A. Begin treatment at once, since the test detects 95% of cases.

B. Ignore the result, since it is probably a false positive.

C. Repeat the same test and treat only if it is negative.

D. Confirm with a second, more accurate test before treatment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q5/5

SET 71 • HSS-MD.B.6 • STATISTICS, INFERENCE, AND DECISIONS

Design and judge fair decision procedures with probabilities

MAIN	Algebra II main question	**HSS-MD.B.6**
BACKWARD 1	Probabilities of equally likely outcomes	**7.SP.C.7.a**
BACKWARD 2	Sample spaces for compound events	**7.SP.C.8.b**
BACKWARD 3	Probability between 0 and 1 and redoing outcomes	**7.SP.C.5**
FORWARD 1	Design fair procedures and count expected trials	**HSS-MD.B.6**
FORWARD 2	Analyze strategies and decisions with probability	**HSS-MD.B.7**

SET 71 • HSS-MD.B.6

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

Three friends choose a driver by flipping a fair coin twice: HH = Ana, HT = Ben, TH = Cy, and TT = flip twice again. Is this fair?

A. No: TT is ignored, so the three probabilities add to 3/4 instead of 1.

B. Yes: each friend has probability 1/4, the same as each of the four outcomes.

C. Yes: HH, HT, and TH are equally likely and TT is redone, so each friend has probability 1/3.

D. No: Ana has probability 1/2, because HH has two heads and the others have only one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q1**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q1/5

Four people use a fair die: 1 = A, 2 = B, 3 = C, 4 = D, and 5 or 6 means roll again. Is this fair?

A. No: D is less likely, because 4 is next to 5 and 6.

B. Yes: each person has probability 1/4 after rerolls.

C. Yes: each person has probability 1/6.

D. No: each person has 1/6, and 1/3 of the time is wasted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q2**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q2/5

A well-mixed hat holds 3 slips with Ana's name and 2 with Ben's. One slip is drawn at random to choose who goes first. Is this fair?

A. No: P(Ana) = 3/5 and P(Ben) = 2/5.

B. Yes: the slips are well mixed.

C. Yes: each name is drawn or not, so each has 1/2.

D. No: P(Ana) = 2/5 and P(Ben) = 3/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q3**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q3/5

A random number generator gives a whole number 1-100, each equally likely. Numbers 1-33 choose A, 34-66 choose B, 67-99 choose C, and 100 means generate again. Is this fair?

A. Yes: each has probability 33/100, and the leftover 1/100 does not matter.

B. Yes: each person has 33 equally likely numbers, and 100 is redone, so each has 1/3.

C. No: A has 33 numbers but C has 34 numbers, so C is a little more likely.

D. No: 100 is wasted, so the probabilities are 0.33 each and do not add to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q4**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q4/5

Two fair dice are rolled; if the sum is even, player 1 wins, otherwise player 2 wins. Is this fair?

A. Yes: 18 of the 36 equally likely outcomes have an even sum, so each player has 1/2.

B. Yes: there are 11 possible sums, and about half of the 11 sums are even.

C. No: there are 6 even sums (2, 4, ..., 12) and only 5 odd sums, so player 1 wins more.

D. No: a sum of 7 is the most likely sum, so player 2 wins more often than player 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 1

Probabilities of equally likely outcomes • Grade 7 • Statistics & Probability • 7.SP.C.7.a

A fair die is rolled. What is P(3)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q1/5

A bag has 3 red and 2 blue slips. What is P(red)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q2/5

True or false?

Each face of a fair die has the same probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q3/5

Two fair coins are flipped, one after the other. P(first is heads and second is tails) is

A. 1/2

B. 1/4

C. 1/8

D. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q4/5

A spinner has 4 equal sections. What is P(one section)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 2

Sample spaces for compound events • Grade 7 • Statistics & Probability • 7.SP.C.8.b

A fair coin is flipped twice. Which list is the sample space of equally likely outcomes?

A. H, T

B. HH, HT, TT

C. two heads, one head, no heads

D. HH, HT, TH, TT

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q1/5

How many outcomes does rolling two dice have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q2/5

True or false?

When a fair coin is flipped twice, the 3 outcomes "two heads," "one head," and "no heads" are equally likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q3/5

How many outcomes for two dice give an even sum?

A. 21

B. 12

C. 18

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q4/5

List the outcomes of rolling two dice with a sum of 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 3

Probability between 0 and 1 and redoing outcomes • Grade 7 • Statistics & Probability • 7.SP.C.5

A fair die is rolled; a roll of 5 or 6 is ignored and the die is rolled again until it shows 1-4. What are the probabilities of the final results 1, 2, 3, and 4?

A. Each becomes 1/5, because one reroll is added.

B. 1 becomes more likely than 4, since it is first.

C. Each stays 1/6, because the die is still fair.

D. They stay equally likely, and each becomes 1/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q1/5

A fair die is rolled, and any 5 or 6 is rerolled until the result is 1-4. What is P(final result is 1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q2/5

True or false?

Probabilities of all outcomes in a sample space add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q3/5

An event that can never happen has probability

A. 1

B. -1

C. 1/2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q4/5

Three equally likely outcomes: what is each probability?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q5/5

SET 71 • HSS-MD.B.6

FORWARD 1

Design fair procedures and count expected trials • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

Which procedure chooses one of 5 people fairly using a fair coin?

A. Flip 5 times; the person whose number equals the count of heads is chosen, and 0 heads means flip again.

B. Flip twice (4 equally likely outcomes); one outcome each for persons 1-4, and person 5 also gets HH.

C. Flip 3 times (8 equally likely outcomes); assign 5 outcomes to the people and flip again on the other 3.

D. Flip 3 times (8 equally likely outcomes); assign 5 outcomes to people and give the other 3 to person 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F1-Q1**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q1/5

Three friends choose a driver by flipping a coin twice: HH, HT, and TH each pick a friend, and TT means flip again. What is the probability that a choice is made on the first try?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions HSS-MD.B.6 • Set 71 • FORWARD 1 • Q2/5

True or false?

A fair procedure must give each person an equal chance, even if it may take several tries.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q3/5

To choose among 6 people fairly with one die, you

A. roll twice and add

B. assign one face to each person

C. assign 1-3 to person A

D. reroll on 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q4/5

A teacher picks a student from a class of 28 using a random number generator that gives whole numbers 1-30, each equally likely. Which rule is fair?

A. Assign numbers 1-28 to the students, and generate again on 29 or 30.

B. Assign numbers 1-28; 29 and 30 both choose student 1.

C. Assign numbers 1-28; 29 and 30 choose students 1 and 2.

D. Generate one number and divide it by 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F1-Q5**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions HSS-MD.B.6 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSS-MD.B.6

FORWARD 2

Analyze strategies and decisions with probability • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A game costs $1. You roll a fair die and win $4 if you roll a 6, and nothing otherwise. Find the expected net gain per game, in dollars, as a fraction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q1/5

A game costs $1. You roll a fair die and win a prize if you roll a 6, and nothing otherwise. With a $4 prize, what is the expected net gain in dollars? What prize makes the expected net gain 0?

Expected net gain ($): _____

Fair prize ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q2/5

True or false?

A game can be fair in who is chosen but still have a negative expected value for the player.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q3/5

With P(win) = 1/4 and prize \$8 for a \$2 ticket, the expected net gain is

A. \$2

B. −\$2

C. \$0

D. \$6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q4/5

What is the difference between a fair choice procedure and a fair game?

A. A fair game always has a prize; a fair choice procedure never uses probability at all.

B. A fair choice procedure has expected net gain 0; a fair game gives everyone equal probability.

C. A fair choice procedure gives everyone equal probability; a fair game has expected net gain 0.

D. They mean the same thing: both require each player to have probability 1/2 of winning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q5/5

SET 72 • HSS-MD.B.7 • STATISTICS, INFERENCE, AND DECISIONS

Analyze decisions and strategies with probabilities

MAIN	Algebra II main question	**HSS-MD.B.7**
BACKWARD 1	Probability of compound independent events	**7.SP.C.8.a**
BACKWARD 2	Two-way tables of counts	**8.SP.A.4**
BACKWARD 3	Weighted averages of outcomes	**6.RP.A.3.c**
FORWARD 1	Investigate how changing conditions changes the decision	**HSS-MD.B.7**
FORWARD 2	Find expected values of random variables	**HSS-MD.A.2**

SET 72 • HSS-MD.B.7

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A disease affects 1% of 10,000 people. A test detects 95% of people who have it and gives a false positive for 5% of healthy people. Find the number of true positives, the number of false positives, and P(disease | positive) as a decimal rounded to the nearest thousandth.

True positives: _____

False positives: _____

P(disease | positive): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q1**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q1/5

A shipment is accepted if 2 randomly tested items are both non-defective. Each item is defective with probability 0.1, independently. What is the probability the shipment is accepted? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q2/5

A $30 warranty covers a repair that costs $200 and is needed with probability 0.1 (otherwise there is no cost). Using expected cost as the decision rule, which is correct?

A. Expected repair cost $0.1 \times 200 = \$20 < \$30$, so the warranty is worth buying.

B. Expected repair cost $0.9 \times 200 = \$180 > \$30$, so the warranty is worth buying.

C. Expected repair cost $200 - 30 = \$170 > \$30$, so the warranty is worth buying.

D. Expected repair cost $0.1 \times 200 = \$20 < \$30$, so the warranty is not worth buying.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q3**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q3/5

Option A pays $10 with probability 0.5 and $0 otherwise; option B pays $50 with probability 0.1 and $0 otherwise. Compare the expected values and the risks.

A. Both have expected value $5; A is riskier, because it pays less.

B. A has expected value $10 and B has $5, so A is better.

C. Both have expected value $5; B is riskier, because it usually pays $0.

D. A has expected value $5 and B has $50, so B is better.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q4**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q4/5

A test gives +1 point for a correct answer, −1/4 point for a wrong answer, and 0 for a blank, on 4-choice questions. Using expected score as the decision rule, should a student with no idea of the answer guess randomly or leave it blank?

A. Leave it blank: the expected score of guessing is (1/4)(1) + (3/4)(−1) = −1/2.

B. Guess: the expected score is (1/4)(1) + (3/4)(−1/4) = 1/16, more than 0 for a blank.

C. Leave it blank: a wrong answer loses points, so guessing always lowers the score.

D. Guess: the expected score is (1/4)(1) = 1/4, because wrong answers do not count.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 1

Probability of compound independent events • Grade 7 • Statistics & Probability • 7.SP.C.8.a

Each item is non-defective with probability 0.9, independently of the others. Find the probability that two items are both non-defective. Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q1/5

Two fair coins: P(both heads)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q2/5

True or false?

For independent events, $P(A \text{ and } B) = P(A) \cdot P(B)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q3/5

Each item is non-defective with probability 0.9, independently of the others. The probability that three items are all non-defective is

A. 0.9

B. 0.27

C. 0.729

D. 2.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q4/5

Each item is defective with probability 0.1, independently of the others. Find the probability that at least one of two items is defective. Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 2

Two-way tables of counts • Grade 8 • Statistics & Probability • 8.SP.A.4

Of 10,000 people, 1% have a disease. How many have it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q1/5

Of 9,900 healthy people, 5% test positive. How many?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q2/5

True or false?

A two-way table can show test results for people with and without a disease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q3/5

If 95 sick and 495 healthy people test positive, the total positives are

A. 10,000

B. 590

C. 95

D. 495

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q4/5

What fraction of positives are sick (95 of 590)? Give a decimal to the nearest thousandth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 3

Weighted averages of outcomes • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Find 10% of $200.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q1/5

Compute 0.5 × 10 and 0.1 × 50.

0.5 × 10 = _____

0.1 × 50 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q2/5

True or false?

$1/4 \times 1 + 3/4 \times (-1/4) = 1/4$.

A. True

B. False

$0.2 \times 30 + 0.8 \times 0$ equals

A. 30

B. 24

C. 0.2

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q4/5

Compute $0.9 \times 0 + 0.1 \times 200$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q5/5

SET 72 • HSS-MD.B.7

FORWARD 1

Investigate how changing conditions changes the decision • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A test detects 95% of people who have a disease and gives a false positive for 5% of healthy people. In a group of 10,000 where 10% have the disease, find P(disease | positive). Give a decimal rounded to the nearest thousandth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q1/5

A $30 warranty covers a repair that costs $200; the repair is needed with probability 0.2 and otherwise there is no cost. Using expected cost as the decision rule, is the warranty worth buying?

A. Yes: the repair cost of $200 is more than $30.

B. No: the repair probably will not happen, since $0.2 < 0.5$.

C. Yes: the expected repair cost is $0.2 \times 200 = \$40$, more than $30.

D. No: the expected repair cost is $40, which is more than $30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q2/5

True or false?

A positive result is more convincing when the disease is more common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q3/5

A test gives +1 point for a correct answer and −1/4 point for a wrong answer on 4-choice questions. A student eliminates one wrong choice and guesses randomly among the 3 remaining choices. What is the expected score for the guess?

A. (1/3)(1) + (2/3)(−1/4) = 1/6

B. (1/3)(1) + (2/3)(−1) = −1/3

C. (1/4)(1) + (3/4)(−1/4) = 1/16

D. (1/3)(1) = 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q4**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q4/5

A shipment is accepted only if all n tested items are non-defective. Each item is defective with probability 0.1, independently. What is the smallest whole number n for which the probability of acceptance is below 0.5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q5/5

SET 72 • HSS-MD.B.7

FORWARD 2

Find expected values of random variables • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

A random variable X takes values 0, 1, 2 with probabilities 0.25, 0.5, 0.25. Find E(X).

E(X) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions HSS-MD.A.2 • Set 72 • FORWARD 2 • Q1/5

A raffle sells 500 tickets at $2 each; one ticket wins a $400 prize. Find the expected net gain of one ticket, in dollars, as a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q2/5

True or false?

The expected value is the long-run average of many repetitions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q3/5

For a fair die, E(roll) is

A. 3

B. 21

C. 3.5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q4**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q4/5

A $30 warranty covers a $200 repair that is needed with probability 0.1 (otherwise no cost). Why is deciding whether to buy it an expected-value calculation?

A. The repair cost is a random variable (0 or 200) with expected value $20, compared with the certain $30 price.

B. The repair will cost exactly $20 for certain, so the warranty price is compared with $20.

C. The expected repair cost is 0.1 + 200 = $200.10, which is compared with the $30 price.

D. The warranty price is a random variable whose expected value is $200, compared with the $30 price.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q5/5

SET 73 • HSN-CN.A.3 • ADVANCED EXTENSION (+)

Use conjugates to find moduli and quotients of complex numbers

MAIN	Algebra II main question	**HSN-CN.A.3**
BACKWARD 1	Multiply complex numbers	**HSN-CN.A.2**
BACKWARD 2	Distance from the origin	**8.G.B.8**
BACKWARD 3	Powers of i	**HSN-CN.A.1**
FORWARD 1	Prove properties of conjugates and moduli	**HSN-CN.A.3**
FORWARD 2	Write complex numbers in polar form	**HSN-CN.B.4**

SET 73 • HSN-CN.A.3

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

Give the conjugate of $3 - 4i$ and find $|3 - 4i|$.

Conjugate: _____

$|3 - 4i|$ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q1/5

Write (2 + 3i)/(1 − i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q2/5

Write (5 + i)/(2 + 3i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q3/5

Find $|-5 + 12i|$ and compute $(-5 + 12i)(-5 - 12i)$.

$|-5 + 12i| =$ _____

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q4/5

Write 1/(4 + 3i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q5/5

SET 73 • HSN-CN.A.3

BACKWARD 1

Multiply complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply $(2 + 3i)(1 + i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q1/5

Multiply $(3 - 4i)(3 + 4i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q2/5

True or false?

$(a + bi)(a - bi) = a^2 + b^2$.

A. True

B. False

$(5 + i)(2 - 3i)$ equals

A. $13 + 13i$

B. $7 - 13i$

C. $10 - 3i$

D. $13 - 13i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q4/5

Multiply $(1 - i)(1 + i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSN-CN.A.3

BACKWARD 2

Distance from the origin • Grade 8 • Geometry • 8.G.B.8

Find the distance from (0, 0) to (3, −4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q1/5

Find the distance from (0, 0) to (−5, 12).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q2/5

True or false?

The distance from (0, 0) to (a, b) is $\sqrt{(a^2 + b^2)}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q3/5

The distance from (0, 0) to (1, 1) is

A. 0

B. 1

C. 2

D. $\sqrt{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q4/5

Find the distance from (0, 0) to (4, 3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSN-CN.A.3

BACKWARD 3

Powers of i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Simplify i^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q1/5

Simplify $-12i \cdot i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q2/5

True or false?

$-3i^2 = -3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q3/5

Which is real?

A. $3i$

B. $(4 + 3i)^2$

C. $(4 + 3i)(4 - 3i)$

D. $4 + 3i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q4/5

Write 13/13 − 13i/13 in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q5/5

SET 73 • HSN-CN.A.3

FORWARD 1

Prove properties of conjugates and moduli • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

Which argument proves $z \cdot \bar{z} = |z|^2$ for $z = a + bi$ (a and b real)?

A. $(a + bi)(a - bi) = a^2 + b^2i^2$, which is $|z|^2$ because $|z| = \sqrt{(a^2 + b^2i^2)}$.

B. $(a + bi)(a - bi) = a^2 - b^2$, and $|z|^2 = a^2 - b^2$.

C. $(a + bi)(a - bi) = a^2 - b^2i^2 = a^2 + b^2$, and $|z|^2 = a^2 + b^2$.

D. For $z = 3 + 4i$: $(3 + 4i)(3 - 4i) = 25 = 5^2$, so the statement holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q1/5

Let $z = 3 + 4i$ and $w = 1 + i$. Find zw, $|zw|$, and $|z||w|$, giving the moduli exactly in simplest radical form.

$zw =$ _____

$|zw| =$ _____

$|z||w| =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q2/5

True or false?

The conjugate of a real number is itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q3/5

If $z = 2 - 5i$, then $z + \bar{z}$ is

A. $4 - 10i$

B. 0

C. $-10i$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q4/5

Use $\bar{z}/|z|^2$ to write $1/z$ for $z = 1 + 2i$ in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q5/5

SET 73 • HSN-CN.A.3

FORWARD 2

Write complex numbers in polar form • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4

Write $1 + i$ in polar form $r(\cos\theta + i\sin\theta)$ with $0 \le \theta < 2\pi$. Give r and θ exactly.

$r =$ _____

$\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q1/5

Write $2(\cos(\pi/2) + i \sin(\pi/2))$ in rectangular form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q2/5

True or false?

The modulus of a complex number is the r in its polar form.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q3/5

The polar form of -3 is

A. $3(\cos \pi + i \sin \pi)$

B. $-3(\cos \pi + i \sin \pi)$

C. $3(\cos 0 + i \sin 0)$

D. $3(\cos(\pi/2) + i \sin(\pi/2))$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q4/5

In the complex plane, what do the modulus and the angle of a complex number represent?

A. The modulus is the sum of the parts a + b; the angle is the ratio of the parts b/a.

B. The modulus is the distance from 1; the angle is measured from the positive imaginary axis.

C. The modulus is the real part; the angle is the imaginary part of the number written as a + bi.

D. The modulus is the distance from 0; the angle is the direction measured from the positive real axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.d • ADVANCED EXTENSION (+)

Graph rational functions with zeros and asymptotes

MAIN	Algebra II main question	**HSF-IF.C.7.d**
BACKWARD 1	Rewrite a rational expression by inspection	**HSA-APR.D.6**
BACKWARD 2	Domain and excluded values	**HSF-IF.B.5**
BACKWARD 3	Sign of a quotient on intervals	**7.NS.A.2**
FORWARD 1	Graph rational functions with slant asymptotes	**HSF-IF.C.7.d**
FORWARD 2	Interpret a rational model's features in context	**HSF-IF.B.4**

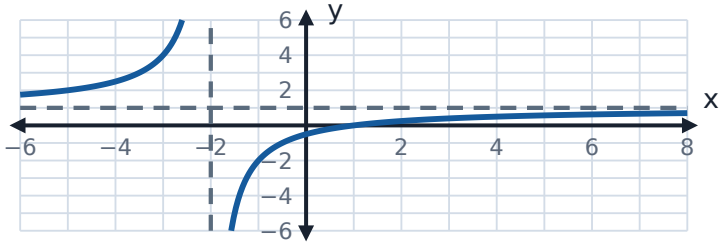
SET 74 • HSF-IF.C.7.d

MAIN

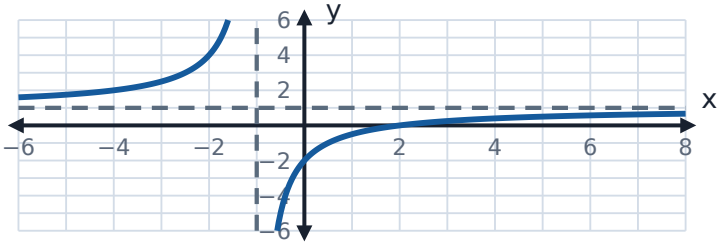
Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Which graph shows y = (x + 1)/(x − 2), with its asymptotes dashed?

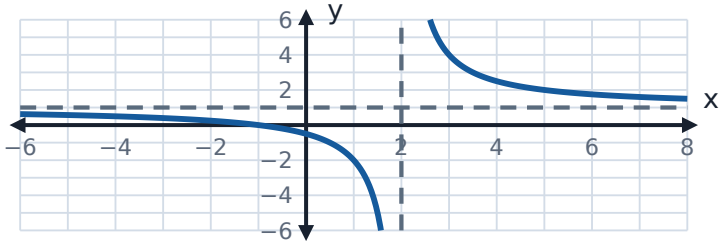
A



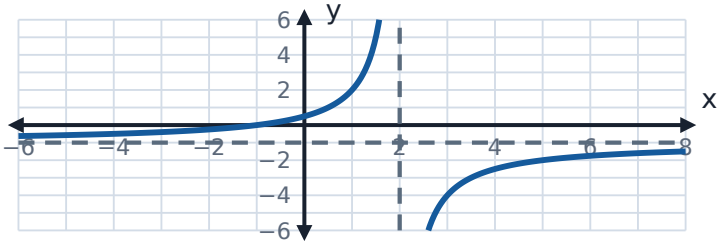
B



C



D



A. Graph A

B. Graph B

C. Graph C

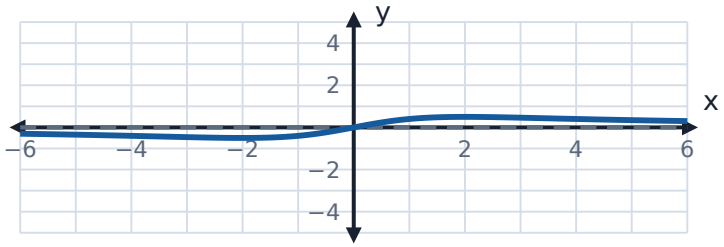
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q1**

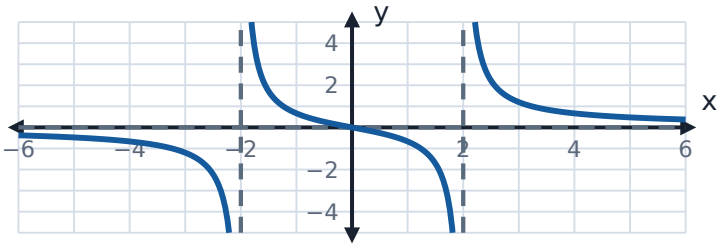
High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q1/5

Which graph shows $y = 2x/(x^2 - 4)$, with its asymptotes dashed?

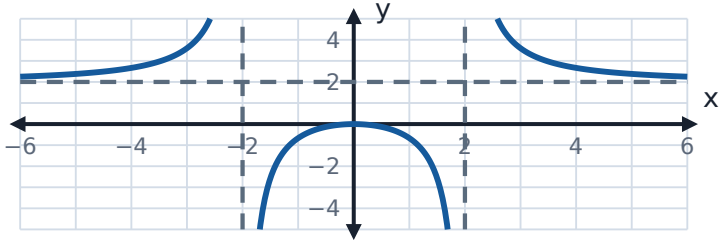
A



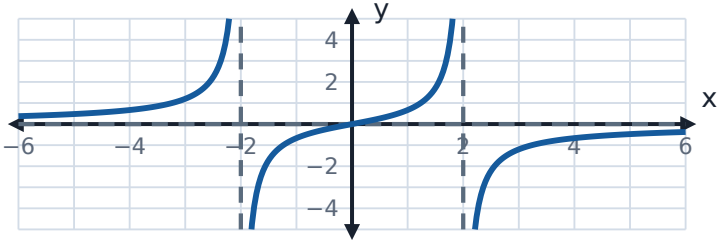
B



C



D



A. Graph A

B. Graph B

C. Graph C

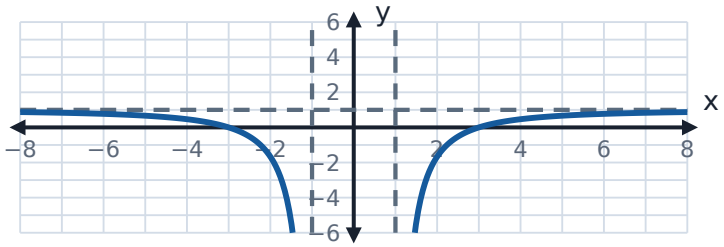
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q2**

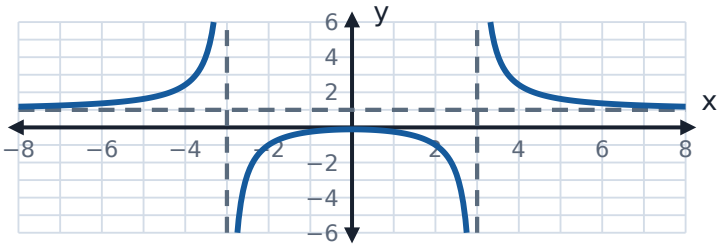
High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q2/5

Which graph shows $y = (x^2 - 1)/(x^2 - 9)$, with its asymptotes dashed?

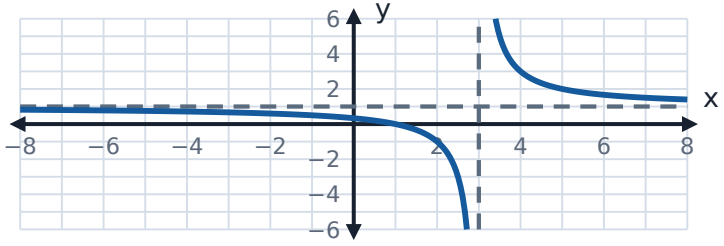
A



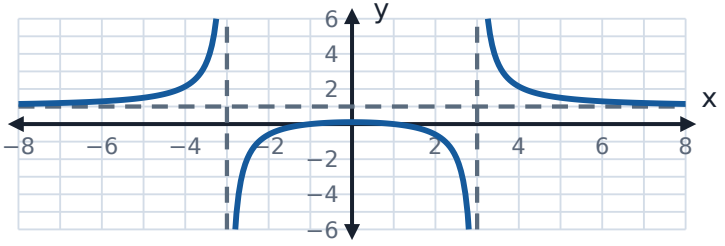
B



C



D



A. Graph A

B. Graph B

C. Graph C

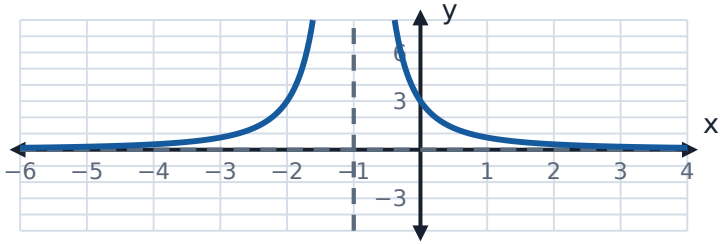
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q3**

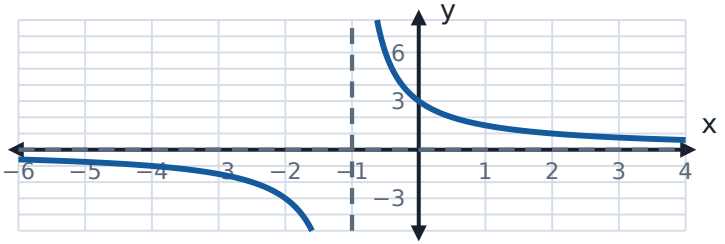
High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q3/5

Which graph shows $y = 3/(x + 1)^2$, with its asymptotes dashed?

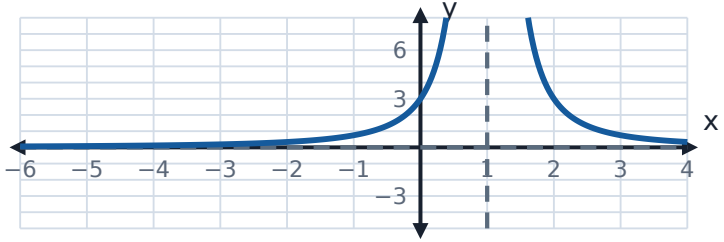
A



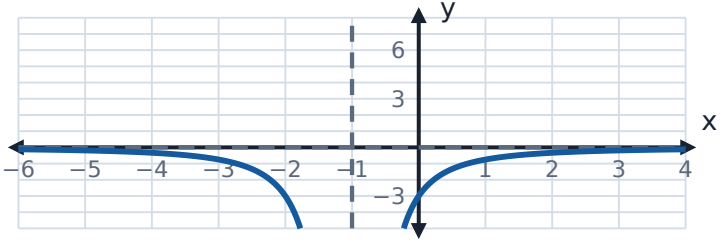
B



C



D



A. Graph A

B. Graph B

C. Graph C

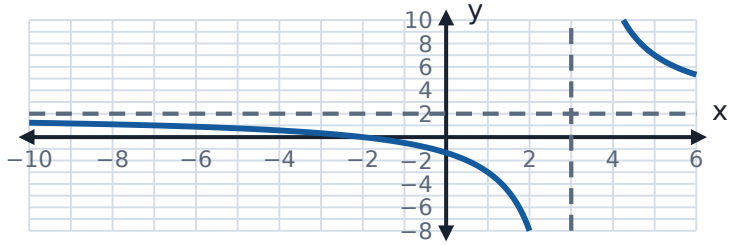
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q4**

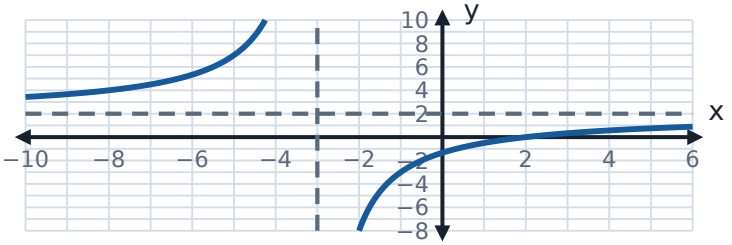
High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q4/5

Which graph shows $y = (2x - 4)/(x + 3)$, with its asymptotes dashed?

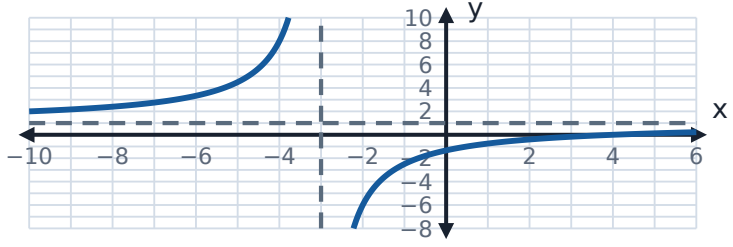
A



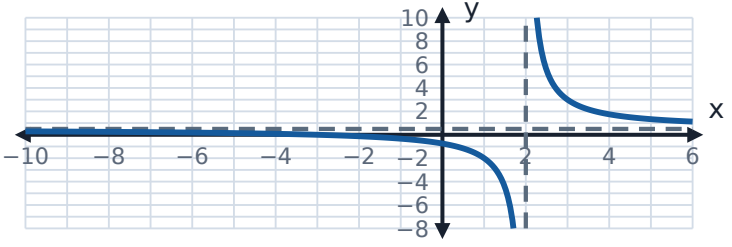
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 1

Rewrite a rational expression by inspection • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Rewrite $(x + 1)/(x - 2)$ as $1 + r/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q1/5

Rewrite $(2x - 4)/(x + 3)$ as $2 + r/(x + 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q2/5

True or false?

The constant q in $q + r/(x - a)$ is the horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q3/5

$(x^2 - 1)/(x^2 - 9)$ equals

A. $1 + 8/(x^2 - 9)$

B. $1 - 8/(x^2 - 9)$

C. $1 + 1/(x^2 - 9)$

D. $1/9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q4/5

Factor the denominator of $2x/(x^2 - 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 2

Domain and excluded values • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

Give the domain of $(x + 1)/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q1/5

Give the domain of $3/(x + 1)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q2/5

True or false?

A rational function is undefined where its denominator is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q3/5

$2x/(x^2 - 4)$ is undefined at

A. $x = \pm 2$

B. $x = 4$

C. $x = 2$ only

D. $x = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q4/5

List the excluded values of $(x^2 - 1)/(x^2 - 9)$, separated by commas.

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 3

Sign of a quotient on intervals • Grade 7 • The Number System • 7.NS.A.2

Is $(x + 1)/(x - 2)$ positive or negative at $x = 0$? Answer Positive or Negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q1/5

Is $2x/(x^2 - 4)$ positive or negative at $x = 3$? Answer Positive or Negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q2/5

True or false?

A quotient of two negative numbers is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q3/5

At $x = 2.1$, $(x + 1)/(x - 2)$ is

A. large and negative

B. undefined

C. zero

D. large and positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q4/5

True or false?

$3/(x + 1)^2$ is negative for some value of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q5/5

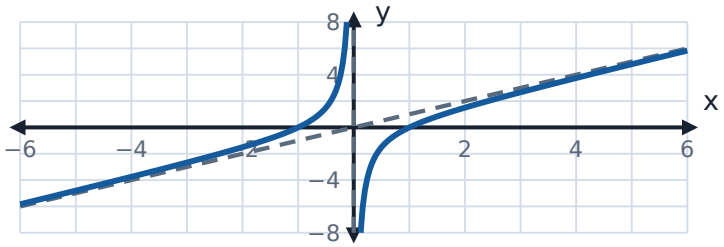
SET 74 • HSF-IF.C.7.d

FORWARD 1

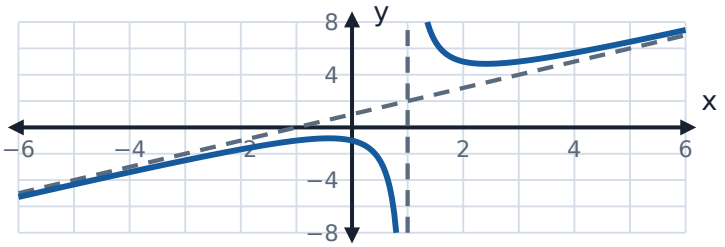
Graph rational functions with slant asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Which graph shows $y = (x^2 + 1)/x$, with its asymptotes dashed?

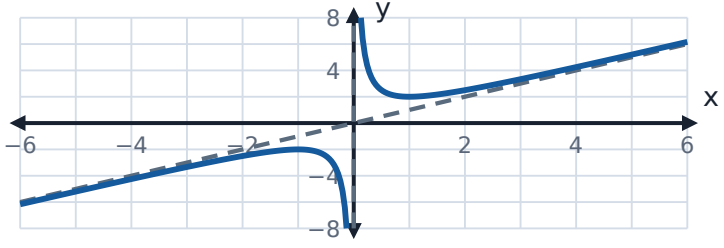
A



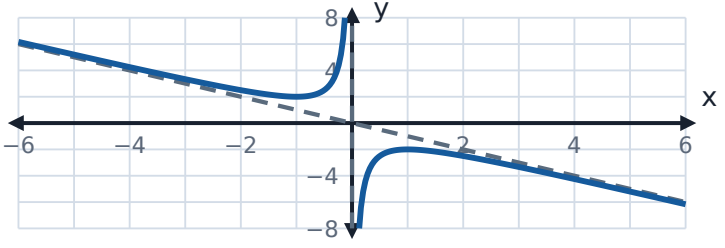
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q1/5

Find the slant asymptote of $y = (x^2 - x - 1)/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q2/5

True or false?

A rational function can cross its horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q3/5

$y = (3x^2 + 1)/(x^2 + 4)$ has horizontal asymptote

A. none

B. $y = 3$

C. $y = 0$

D. $y = 1/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q4/5

Which rational function has vertical asymptote $x = 1$, horizontal asymptote $y = 2$, and a zero at $x = 3$?

A. $f(x) = (x - 3)/(2(x - 1))$

B. $f(x) = 2(x - 3)/(x - 1)$

C. $f(x) = 2(x - 1)/(x - 3)$

D. $f(x) = 2(x + 3)/(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q5/5

SET 74 • HSF-IF.C.7.d

FORWARD 2

Interpret a rational model's features in context • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

The average cost per item is $A(x) = (500 + 4x)/x$ dollars for x items. What do the vertical asymptote $x = 0$ and the horizontal asymptote $y = 4$ mean in context?

A. $x = 0$: the fixed cost of production is \$0; $y = 4$: no more than 4 items can be made at this average cost per item.

B. $x = 0$: the average cost grows without bound as the number of items nears 0; $y = 4$: the average cost approaches \$4 per item as x grows.

C. $x = 0$: the first item is free to make; $y = 4$: the total cost of all of the items approaches \$4 as the number of items grows.

D. $x = 0$: the average cost is exactly \$500 for very few items; $y = 4$: the average cost is never more than \$4 per item.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q1/5

A drug concentration is $C(t) = 20t/(t^2 + 4)$ for $t \geq 0$ hours. What does its horizontal asymptote mean?

A. $y = 0$: the concentration is 0 at every time after $t = 2$.

B. $y = 20$: the concentration approaches 20 as time goes on.

C. $y = 0$: the drug never enters the blood.

D. $y = 0$: the concentration approaches 0 as time goes on.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q2/5

True or false?

For a context, only the part of a rational graph in the context domain is meaningful.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q3/5

In $A(x) = (500 + 4x)/x$, as x grows the average cost

A. increases

B. is constant

C. approaches 0

D. decreases toward 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q4/5

The average cost per item is $A(x) = (500 + 4x)/x$ dollars for x items. Find the number of items for which the average cost is $5.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q5/5

SET 75 • HSF-IF.C.7.d • ADVANCED EXTENSION (+)

Distinguish holes from vertical asymptotes

MAIN	Algebra II main question	**HSF-IF.C.7.d**
BACKWARD 1	Factor numerators and denominators	**HSA-SSE.A.2**
BACKWARD 2	Simplify rational expressions and keep exclusions	**HSA-APR.D.7**
BACKWARD 3	Evaluate the simplified expression	**HSF-IF.A.2**
FORWARD 1	Construct and graph rational functions with holes	**HSF-IF.C.7.d**
FORWARD 2	Relate holes to extraneous solutions	**HSA-REI.A.2**

SET 75 • HSF-IF.C.7.d

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

$f(x) = (x^2 - 4)/(x - 2)$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q1/5

$f(x) = (x - 3)/(x^2 - 9)$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q2/5

$f(x) = (x^2 + x - 6)/(x^2 - x - 2)$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q3/5

$f(x) = (x + 1)/(x^2 - 1)$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q4/5

$f(x) = x(x - 1)/(x(x + 2))$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q5/5

SET 75 • HSF-IF.C.7.d

BACKWARD 1

Factor numerators and denominators • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 + x - 6$ and $x^2 - x - 2$.

$x^2 + x - 6 =$ _____

$x^2 - x - 2 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q1/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q2/5

True or false?

$x^2 - 1 = (x - 1)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q3/5

$x^2 - 4$ equals

A. $(x - 4)(x + 1)$

B. $(x - 2)^2$

C. $(x - 2)(x + 2)$

D. $x(x - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q4/5

Factor $x^2 + 2x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q5/5

SET 75 • HSF-IF.C.7.d

BACKWARD 2

Simplify rational expressions and keep exclusions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Simplify $(x - 3)/((x - 3)(x + 3))$ and list the excluded values of the original expression, separated by commas.

Simplified: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q1/5

Simplify $(x + 1)/((x - 1)(x + 1))$ and list the excluded values of the original expression, separated by commas.

Simplified: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q2/5

True or false?

After simplifying, the original excluded values are still excluded.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q3/5

$x(x - 1)/(x(x + 2))$ simplifies to

A. $-1/2$

B. $(x - 1)/(x + 2)$ for all x

C. $x - 1$

D. $(x - 1)/(x + 2)$, $x \neq 0, -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q4/5

Simplify $(x^2 - 4)/(x - 2)$ and give the excluded value of the original expression.

Simplified: _____

Excluded value: $x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q5/5

SET 75 • HSF-IF.C.7.d

BACKWARD 3

Evaluate the simplified expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Evaluate 1/(x + 3) at x = 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q1/5

Evaluate $(x + 3)/(x + 1)$ at $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q2/5

True or false?

The y-coordinate of a hole is the value of the simplified expression there.

A. True

B. False

$(x - 1)/(x + 2)$ at $x = 0$ is

A. 1/2

B. $-1/2$

C. undefined

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q4/5

Evaluate $1/(x - 1)$ at $x = -1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q5/5

SET 75 • HSF-IF.C.7.d

FORWARD 1

Construct and graph rational functions with holes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Which rational function has a hole at $x = 1$, a vertical asymptote $x = -3$, and horizontal asymptote $y = 0$?

A. $f(x) = (x - 1)/((x - 1)(x + 3))$

B. $f(x) = (x - 1)(x + 3)/(x - 1)$

C. $f(x) = (x - 1)/(x + 3)$

D. $f(x) = (x + 3)/((x - 1)(x + 3))$

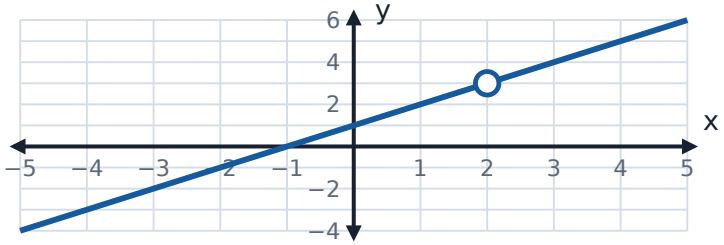
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q1

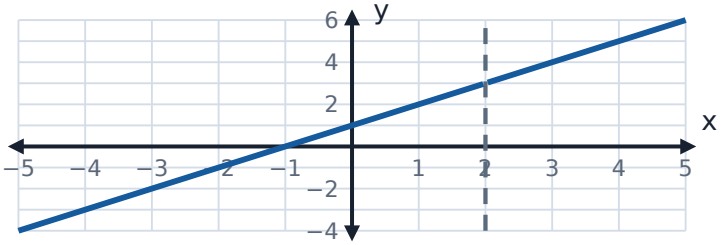
High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q1/5

Which graph shows $y = (x^2 - x - 2)/(x - 2)$?

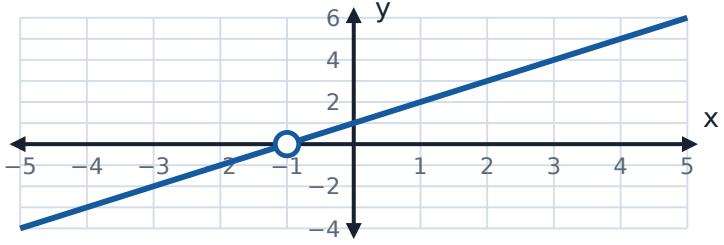
A



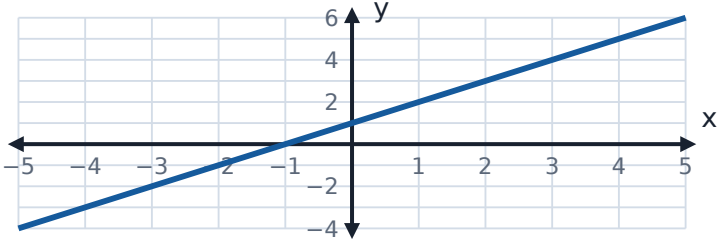
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q2/5

True or false?

A rational function can have both a hole and a vertical asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q3/5

$f(x) = (x - 4)^2/(x - 4)$ has

A. no special features

B. a hole at (4, 4)

C. a vertical asymptote $x = 4$

D. a hole at (4, 0)

For $f(x) = (x - 3)/(x^2 - 9)$, why is $x = -3$ a vertical asymptote but $x = 3$ a hole?

A. Negative excluded values always give asymptotes, and positive excluded values always give holes in a graph.

B. Both $x - 3$ and $x + 3$ cancel, so $x = -3$ and $x = 3$ are both holes, and the asymptote comes from x^2 alone.

C. After canceling $x - 3$, the factor $x + 3$ remains in the denominator, but $x - 3$ does not; near 3 the values approach 1/6.

D. The numerator is 0 at $x = -3$, so the graph has a vertical asymptote at -3, and a hole is at the other zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-F1-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q5/5

SET 75 • HSF-IF.C.7.d

FORWARD 2

Relate holes to extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $(x^2 - 4)/(x - 2) = 4$.

A. $x = 4$, because the hole of the graph is at the height $y = 4$.

B. $x = 2$, because it satisfies the simplified equation $x + 2 = 4$.

C. Every $x \neq 2$, because the function equals 4 at every point of its domain.

D. No solution: $x + 2 = 4$ gives $x = 2$, which is excluded (the hole at (2, 4)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q1/5

Solve $(x - 3)/(x^2 - 9) = 1/6$.

A. $x = 3$ or $x = -3$: both make $x^2 - 9 = 0$, so both solve it.

B. $x = -3$: then $1/(x + 3) = 1/6$ after simplifying the left side.

C. No solution: $1/(x + 3) = 1/6$ gives $x = 3$, which is excluded.

D. $x = 3$: it satisfies $1/(x + 3) = 1/6$, the simplified equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q2/5

True or false?

An extraneous solution of f(x) = c can occur exactly at a hole whose height is c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q3/5

$(x + 1)/(x^2 - 1) = -1/2$ has

A. no solution

B. $x = -1$

C. $x = 0$

D. $x = 1$

Solve $(x^2 + x - 6)/(x^2 - x - 2) = 2$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q5/5

SET 76 • HSF-BF.A.1.c • ADVANCED EXTENSION (+)

Compose functions and interpret compositions with domains

MAIN	Algebra II main question	**HSF-BF.A.1.c**
BACKWARD 1	Build combined models	**HSF-BF.A.1.b**
BACKWARD 2	Evaluate a function at an expression	**HSF-IF.A.2**
BACKWARD 3	Domains of radical, rational, and log functions	**HSF-IF.B.5**
FORWARD 1	Decompose functions into compositions	**HSF-BF.A.1.c**
FORWARD 2	Verify inverses by composition	**HSF-BF.B.4.b**

SET 76 • HSF-BF.A.1.c

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

$f(x) = \sqrt{x}$ and $g(x) = x - 4$. Which gives $f(g(x))$ and $g(f(x))$ with the domain of each?

A. $f(g(x)) = \sqrt{x - 4}$, $x \geq 0$; $g(f(x)) = \sqrt{(x - 4)}$, $x \geq 4$

B. $f(g(x)) = \sqrt{(x - 4)}$, $x \geq 4$; $g(f(x)) = \sqrt{x} - 4$, $x \geq 0$

C. $f(g(x)) = \sqrt{(x - 4)}$, all real x; $g(f(x)) = \sqrt{x} - 4$, all real x

D. $f(g(x)) = \sqrt{(x - 4)}$, $x \geq 0$; $g(f(x)) = \sqrt{x} - 4$, $x \geq 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q1/5

The temperature is $T(h) = 20 - 6.5h$ °C at height h km, and a balloon's height is $h(t) = 0.3t$ km after t minutes. Write $T(h(t)) = a + bt$, and find the temperature at $t = 10$.

$a =$ _____

$b =$ _____

Temperature at $t = 10$ (°C): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q2/5

$f(x) = 1/(x - 1)$ and $g(x) = x^2$. Write $f(g(x))$ and list the real numbers excluded from its domain, separated by commas. Use ^ for exponents.

$f(g(x)) =$ _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q3/5

A dollar amount d converts to euros by e(d) = 0.9d, and a store's price with tax in euros is p(e) = 1.2e. Then p(e(d)) = kd. Find k and the price in euros for d = 50.

k = _____

p(e(50)) (euros): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q4/5

$f(x) = \log_2 x$ and $g(x) = x^2 - 9$. Which gives $f(g(x))$ and its domain?

A. $f(g(x)) = \log_2(x^2 - 9)$; domain $x < -3$ or $x > 3$

B. $f(g(x)) = (\log_2 x)^2 - 9$; domain $x > 0$

C. $f(g(x)) = \log_2(x^2) - 9$; domain $x \neq 0$

D. $f(g(x)) = \log_2(x^2 - 9)$; domain $x > 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 1

Build combined models • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

$f(x) = x^2$ and $g(x) = x + 1$. Find $f(x) + g(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q1/5

$f(x) = x^2$ and $g(x) = x + 1$. Find $f(x) \cdot g(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q2/5

True or false?

f(g(x)) is a different combination from f(x) · g(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q3/5

For $f(x) = 2x$ and $g(x) = x + 3$, $f(x) - g(x)$ is

A. $x - 3$

B. $2x^2 + 6x$

C. $x + 3$

D. $3x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q4/5

Which operation combines T and h in T(h(t))?

A. division

B. addition

C. composition

D. multiplication

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 2

Evaluate a function at an expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$f(x) = \sqrt{x}$. Find $f(x - 4)$.

A. $\sqrt{x} - \sqrt{4}$

B. $\sqrt{(x)} \cdot (-4)$

C. $\sqrt{x} - 4$

D. $\sqrt{(x - 4)}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q1/5

$T(h) = 20 - 6.5h$. Find $T(0.3t)$ in the form $a + bt$. Give a and b.

$a =$ _____

$b =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q2/5

True or false?

If $f(x) = 1/(x - 1)$, then $f(x^2) = 1/(x^2 - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q3/5

If $g(x) = x^2 - 9$, then $g(3)$ is

A. -3

B. -6

C. 9

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q4/5

p(e) = 1.2e. Find p(45).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 3

Domains of radical, rational, and log functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

Give the domain of $\sqrt{(x - 4)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q1/5

Give the domain of $\log_2 u$ in terms of u.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q2/5

True or false?

$1/(x^2 - 1)$ is undefined only at $x = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q3/5

$x^2 - 9 > 0$ when

A. $x < -3$ or $x > 3$

B. $-3 < x < 3$

C. all x

D. $x > 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q4/5

Give the domain of $\sqrt{x} - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q5/5

SET 76 • HSF-BF.A.1.c

FORWARD 1

Decompose functions into compositions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

Which pair writes $h(x) = (3x + 1)^5$ as $f(g(x))$?

A. $g(x) = 3x + 1$, $f(x) = x^5$

B. $g(x) = 3x + 1$, $f(x) = 5x$

C. $g(x) = 3x$, $f(x) = x^5 + 1$

D. $g(x) = x^5$, $f(x) = 3x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q1/5

Which pair writes $h(x) = 2^{x^2}$ as $f(g(x))$?

A. $g(x) = 2x$, $f(x) = x^2$

B. $g(x) = x^2$, $f(x) = 2^x$

C. $g(x) = x^2$, $f(x) = 2x$

D. $g(x) = 2^x$, $f(x) = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q2/5

True or false?

$h(x) = (x + 1)^2$ can be written as $f(g(x))$ in more than one way, for example with $g(x) = x + 1$ and $f(x) = x^2$, or with $g(x) = x^2 + 2x$ and $f(x) = x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q3/5

If $f(x) = x + 1$, $g(x) = 2x$, $k(x) = x^2$, then $f(g(k(3)))$ is

A. 37

B. 19

C. 64

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q4/5

Which context describes a composition V(r(t))?

A. The time t depends on the volume V, and the volume depends on the radius r.

B. The volume V of a balloon depends on its radius r, and the radius depends on time t.

C. The volume V of a balloon is multiplied by its radius r at each time t.

D. The volume V and the radius r of a balloon are added together at each time t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSF-BF.A.1.c

FORWARD 2

Verify inverses by composition • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

Which work shows that $f(x) = 2x + 3$ and $g(x) = (x - 3)/2$ are inverses?

A. $f(0) = 3$ and $g(0) = -3/2$, which are opposite in sign

B. $f(x) \cdot g(x) = (2x + 3)(x - 3)/2 = x$

C. $f(g(x)) = 2((x - 3)/2) + 3 = x + 6$ and $g(f(x)) = x$

D. $f(g(x)) = 2((x - 3)/2) + 3 = x$ and $g(f(x)) = ((2x + 3) - 3)/2 = x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q1/5

Compute $f(g(x))$ for $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q2/5

True or false?

If f(g(x)) = x for all x in g's domain and g(f(x)) = x for all x in f's domain, then f and g are inverses.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q3/5

For $f(x) = x + 5$ and $g(x) = x - 5$, $f(g(2))$ is

A. 2

B. -3

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q4/5

Why is composition the natural test for inverse functions?

A. Composition adds the two functions, and inverses must add to 0 for every input x.

B. Composition reverses the graph across the x-axis, which is where an inverse must lie.

C. An inverse undoes the function, so applying one after the other must return the original input.

D. Two functions are inverses exactly when their product f(x) · g(x) is equal to 1 for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q5/5

SET 77 • HSF-BF.B.4.b • ADVANCED EXTENSION (+)

Verify inverses by composition on the stated domains

MAIN	Algebra II main question	**HSF-BF.B.4.b**
BACKWARD 1	Find inverses algebraically	**HSF-BF.B.4.a**
BACKWARD 2	Compose functions	**HSF-BF.A.1.c**
BACKWARD 3	Square roots of squares	**HSN-RN.A.2**
FORWARD 1	Show that proposed inverses fail	**HSF-BF.B.4.b**
FORWARD 2	Read inverse values from tables and graphs	**HSF-BF.B.4.c**

SET 77 • HSF-BF.B.4.b

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

Which work verifies by composition that $f(x) = 2x + 3$ and $g(x) = (x - 3)/2$ are inverses?

A. $f(x) \cdot g(x) = (2x + 3)(x - 3)/2 = 1$, so f and g are inverses.

B. $f(g(x)) = 2 \cdot (x - 3)/2 + 3 = x$ and $g(f(x)) = ((2x + 3) - 3)/2 = x$, for all real x.

C. $f(1) = 5$ and $g(5) = 1$, so f and g are inverses.

D. $f(g(x)) = 2 \cdot (x - 3)/2 + 3 = x$ for all real x, and one composition is enough by itself.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q1/5

Which work verifies by composition that $f(x) = x^3 - 1$ and $g(x) = \sqrt[3]{(x + 1)}$ are inverses?

A. $f(g(x)) = (\sqrt[3]{(x + 1)})^3 - 1 = x$, but only for $x \geq 0$, so f and g are inverses only when $x \geq 0$.

B. $g(f(x)) = \sqrt[3]{(x^3 - 1)} + 1 = x$, so f and g are inverses.

C. $f(2) = 7$ and $g(7) = 2$, so f and g are inverses.

D. $f(g(x)) = (\sqrt[3]{(x + 1)})^3 - 1 = x$ and $g(f(x)) = \sqrt[3]{((x^3 - 1) + 1)} = \sqrt[3]{(x^3)} = x$, for all real x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q2/5

Which work verifies by composition that $f(x) = 2^x$ and $g(x) = \log_2 x$ are inverses, with the correct domains?

A. $f(g(x)) = 2 \cdot \log_2 x = x$ for $x > 0$.

B. $f(g(x)) = 2^{\log_2 x} = x$ and $g(f(x)) = \log_2(2^x) = x$, both for every real number x.

C. $f(3) = 8$ and $g(8) = 3$, so f and g are inverses for all x.

D. $f(g(x)) = 2^{\log_2 x} = x$ for $x > 0$, and $g(f(x)) = \log_2(2^x) = x$ for all real x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q3/5

Which work verifies by composition that $f(x) = 1/(x - 2)$ $(x \neq 2)$ and $g(x) = 1/x + 2$ $(x \neq 0)$ are inverses?

A. $f(3) = 1$ and $g(1) = 3$, so f and g are inverses.

B. $f(g(x)) = 1/((1/x + 2) - 2) = 1/(1/x) = x$, so f and g are inverses for all real x with no excluded values needed.

C. $f(g(x)) = 1/((1/x + 2) - 2) = 1/(1/x) = x$ for $x \neq 0$, and $g(f(x)) = 1/(1/(x - 2)) + 2 = (x - 2) + 2 = x$ for $x \neq 2$.

D. $g(f(x)) = 1/(x - 2) + 2 = (2x - 3)/(x - 2)$, so they are not inverses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q4/5

$f(x) = x^2$ with domain $x \geq 0$ and $g(x) = \sqrt{x}$. Which work verifies that they are inverses and explains why the domain restriction is needed?

A. The restriction is needed because $\sqrt{x}$ is undefined for $x > 0$.

B. $f(g(x)) = (\sqrt{x})^2 = x$ for $x \geq 0$, and $g(f(x)) = \sqrt{(x^2)} = |x| = x$ because $x \geq 0$. Without the restriction, $\sqrt{(x^2)} = |x| \neq x$ for $x < 0$.

C. $f(g(x)) = (\sqrt{x})^2 = x$ and $g(f(x)) = \sqrt{(x^2)} = x$ for every real number x, so the two functions are inverses and no domain restriction is needed.

D. $g(f(x)) = \sqrt{(x^2)} = \pm x$, and the restriction chooses the + sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 1

Find inverses algebraically • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

Find the inverse of $f(x) = 2x + 3$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q1/5

Find the inverse of $f(x) = x^3 - 1$.

A. $f^{-1}(x) = \sqrt[3]{(x + 1)}$

B. $f^{-1}(x) = \sqrt[3]{x} + 1$

C. $f^{-1}(x) = \sqrt[3]{(x - 1)}$

D. $f^{-1}(x) = 1/(x^3 - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q2/5

True or false?

The inverse of 2^x is $\log_2 x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q3/5

The inverse of $f(x) = 1/(x - 2)$ is

A. $1/x + 2$

B. $1/(x + 2)$

C. $2 - 1/x$

D. $x - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q4/5

Find the inverse of $f(x) = x^2$ for $x \geq 0$.

A. $f^{-1}(x) = \pm\sqrt{x}$

B. $f^{-1}(x) = \sqrt{x}$

C. $f^{-1}(x) = x/2$

D. $f^{-1}(x) = 1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 2

Compose functions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

$f(x) = 2x + 3$ and $g(x) = (x - 3)/2$. Compute $f(g(5))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q1/5

$f(x) = 2x + 3$ and $g(x) = (x - 3)/2$. Compute $g(f(5))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q2/5

True or false?

f(g(x)) means apply g first, then f.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q3/5

For $f(x) = x^3$ and $g(x) = x + 1$, $f(g(x))$ is

A. $(x + 1)^3$

B. $x^3 + 1$

C. $x + 1$

D. $3x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q4/5

Compute $\log_2(2^5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 3

Square roots of squares • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Evaluate $\sqrt{((-3)^2)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q1/5

Evaluate $(\sqrt{7})^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{(x^2)} = x$ for every real x.

A. True

B. False

$\sqrt{(x^2)}$ equals

A. $|x|$

B. x^2

C. $-x$

D. x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q4/5

For which x is $\sqrt{(x^2)} = x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q5/5

SET 77 • HSF-BF.B.4.b

FORWARD 1

Show that proposed inverses fail • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

Which work shows by composition that $f(x) = 3x - 1$ and $g(x) = (x - 1)/3$ are not inverses?

A. $f(0) = -1$ and $g(0) = -1/3$ are different, so they are not inverses

B. $f(g(x)) = 3((x - 1)/3) - 1 = x$, so they are inverses

C. $g(f(x)) = ((3x - 1) - 1)/3 = x$, so they are inverses

D. $f(g(x)) = 3((x - 1)/3) - 1 = x - 2$, which is not x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q1/5

$f(x) = x^2$ (all real x) and $g(x) = \sqrt{x}$. Which composition fails to equal x, and for which x?

A. $g(f(x)) = \sqrt{(x^2)} = |x|$, which is not x for $x < 0$

B. $f(g(x)) = (\sqrt{x})^2$ fails for every $x > 0$

C. $g(f(x)) = x$ for every real x

D. $f(g(x)) = |x|$, which is not x for $x < 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q2/5

True or false?

Checking f(g(x)) = x alone is always enough.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q3/5

For $f(x) = e^x$ and $g(x) = \ln x$, $g(f(x)) =$

A. $|x|$

B. x only for $x > 0$

C. x for all real x

D. $e^{\ln x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q4/5

Fix the domain of $f(x) = (x - 1)^2$ so that $g(x) = 1 + \sqrt{x}$ is its inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSF-BF.B.4.b

FORWARD 2

Read inverse values from tables and graphs • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. Find $f^{-1}(7)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q1/5

A table gives $f(0) = 1$, $f(1) = 3$, $f(2) = 7$, $f(3) = 15$. Find $f^{-1}(f^{-1}(15))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q2/5

True or false?

If (a, b) is on the graph of f, then (b, a) is on the graph of f^{-1}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q3/5

If $f(4) = 9$, then $f^{-1}(9)$ is

A. 1/4

B. 4

C. 13

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q4/5

A table gives $f(1) = 3$ and $f(2) = 7$. How do these values give $f(f^{-1}(7)) = 7$?

A. $f^{-1}(7) = 1/7$ because an inverse is a reciprocal, and $f(1/7) = 7$, so $f(f^{-1}(7)) = 7$.

B. $f^{-1}(7) = 7 - 3 = 4$ by subtracting the outputs, and $f(4) = 7$ by the pattern.

C. $f^{-1}(7) = 2$ because $f(2) = 7$, and then $f(2) = 7$, so $f(f^{-1}(7)) = 7$.

D. $f^{-1}(7) = 3$ because $f(1) = 3$, and then $f(3) = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q5/5

SET 78 • HSF-BF.B.4.c • ADVANCED EXTENSION (+)

Read values of an inverse function from graphs and tables

MAIN	Algebra II main question	**HSF-BF.B.4.c**
BACKWARD 1	Inverse functions swap inputs and outputs	**HSF-BF.B.4.a**
BACKWARD 2	Read points from graphs and tables	**8.F.A.1**
BACKWARD 3	One-to-one functions	**HSF-BF.B.4.b**
FORWARD 1	Estimate inverse values between data points and sketch inverse...	**HSF-BF.B.4.c**
FORWARD 2	Restrict the domain to make a function invertible	**HSF-BF.B.4.d**

SET 78 • HSF-BF.B.4.c

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

The table shows f. Find $f^{-1}(15)$.

x	0	1	2	3	4
f(x)	1	3	7	15	31

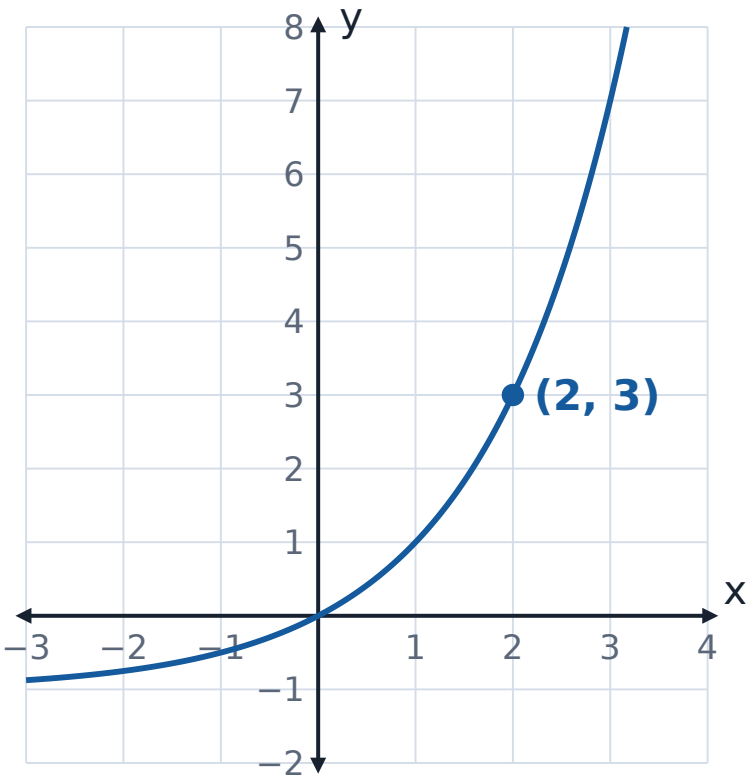
$f^{-1}(15) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q1/5

The graph of $f(x) = 2^x - 1$ is shown. Use it to find $f^{-1}(3)$.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q2/5

The table shows a decreasing function g. Find $g^{-1}(10)$ and $g^{-1}(g(2))$.

x	1	2	3	4	5
g(x)	40	25	16	10	4

$g^{-1}(10) =$ _____

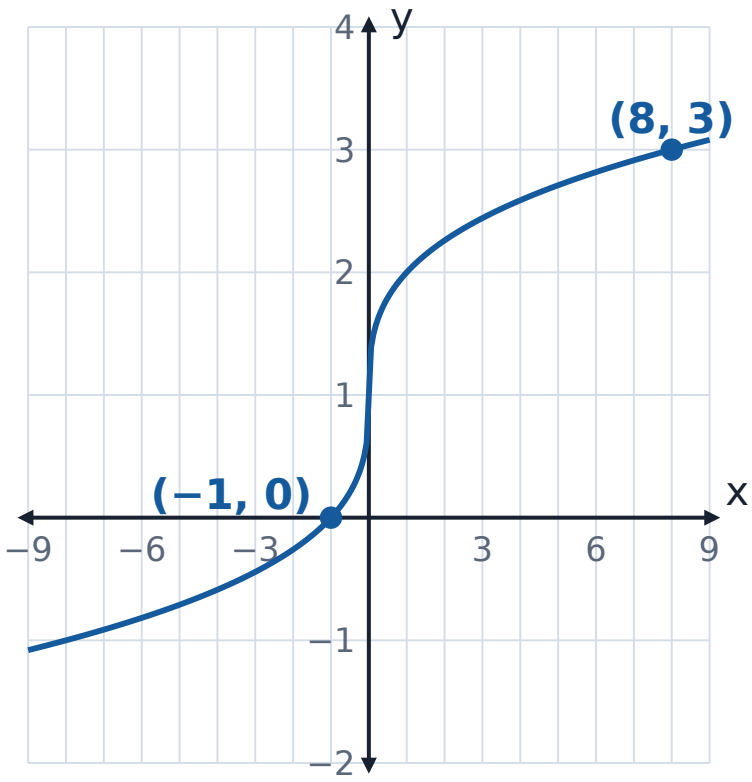
$g^{-1}(g(2)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q3/5

The graph of $f(x) = \sqrt[3]{x} + 1$ is shown. Use it to find $f^{-1}(0)$ and $f^{-1}(3)$.



$f^{-1}(0) =$ _____

$f^{-1}(3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q4/5

The table gives a plane's altitude A (m) t minutes after takeoff. Find $A^{-1}(1200)$ and interpret it.

t (min)	0	1	2	3	4	5
A (m)	0	250	550	880	1200	1500

A. $A^{-1}(1200) = 4$: the plane climbs 1200 m every minute.

B. $A^{-1}(1200) = 1500$: the altitude one minute after reaching 1200 m.

C. $A^{-1}(1200) = 4$: the plane reaches 1200 m four minutes after takeoff.

D. $A^{-1}(1200) = 1/1200$: the reciprocal of the altitude.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 1

Inverse functions swap inputs and outputs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

If $f(3) = 15$, what is $f^{-1}(15)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q1/5

Write the inverse of $f(x) = 2^x - 1$.

A. $f^{-1}(x) = \log_2 x + 1$

B. $f^{-1}(x) = 2^{-x} + 1$

C. $f^{-1}(x) = \log_2(x - 1)$

D. $f^{-1}(x) = \log_2(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q2/5

True or false?

$f^{-1}(y)$ is the input of f that gives the output y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q3/5

If (5, 4) is on the graph of g, which point is on g^{-1}?

A. (5, 4)

B. (1/5, 1/4)

C. (−5, −4)

D. (4, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q4/5

Use the formula to find $f^{-1}(3)$ for $f(x) = 2^x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 2

Read points from graphs and tables • Grade 8 • Functions • 8.F.A.1

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. What is f(2)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q1/5

A plane's altitude A (m) at t = 0, 1, 2, 3, 4 minutes is 0, 250, 550, 880, 1200. What is A(3)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q2/5

True or false?

A function gives exactly one output for each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q3/5

$g(1) = 40$, $g(2) = 25$, $g(3) = 16$, $g(4) = 10$. Which input of g gives 16?

A. 16

B. 3

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q4/5

A graph passes through (8, 3). What is the output at input 8?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 3

One-to-one functions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

Why does a strictly increasing function have an inverse function?

A. Its graph passes through the origin, so it can be reflected over $y = x$.

B. Different inputs give different outputs, so each output comes from exactly one input.

C. Its outputs are always positive, so each output can be undone.

D. Its domain is all real numbers, so every input has an output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q1/5

Does a function with f(1) = 4 and f(3) = 4 have an inverse function? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q2/5

True or false?

A decreasing function can have an inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q3/5

Which table has an inverse function?

A. x: 1, 2, 3; y: 5, 7, 5

B. x: 1, 2, 3; y: 5, 7, 9

C. x: 1, 2, 3; y: 0, 1, 0

D. x: 1, 2, 3; y: 2, 2, 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q4/5

A table gives $f(2) = 7$ and $f(3) = 15$. Use the table to find $f^{-1}(15)$, then $f(f^{-1}(15))$.

$f^{-1}(15) =$ _____

$f(f^{-1}(15)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q5/5

SET 78 • HSF-BF.B.4.c

FORWARD 1

Estimate inverse values between data points and sketch inverse graphs • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

A plane's altitude is 880 m at t = 3 min and 1200 m at t = 4 min. Assume it climbs at a constant rate between these times. Estimate the time when it reaches 1000 m, to the nearest tenth of a minute.

t ≈ _____

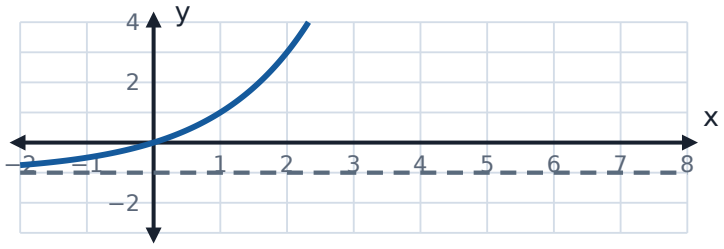
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q1

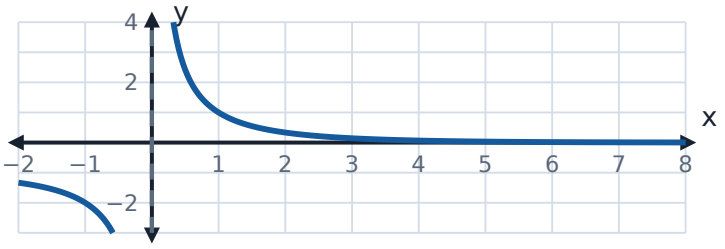
High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q1/5

Which graph shows the inverse of $f(x) = 2^x - 1$?

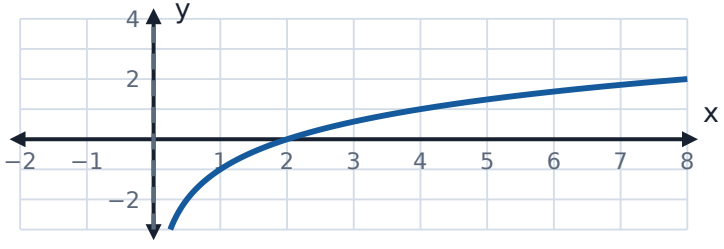
A



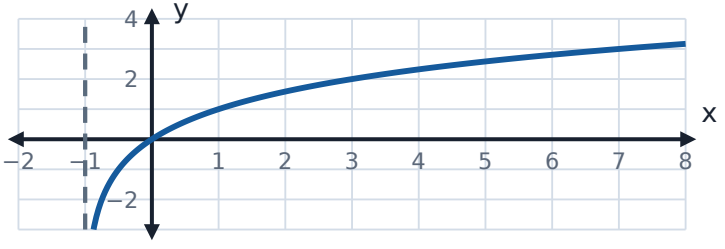
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S78-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q2/5

True or false?

The graph of f^{-1} is the reflection of the graph of f across $y = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q3/5

A decreasing function has $g(1) = 40$, $g(2) = 25$, $g(3) = 16$, $g(4) = 10$. Estimate $g^{-1}(20)$.

A. between 1 and 2

B. between 2 and 3

C. between 4 and 5

D. between 3 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q4/5

A table gives $f(0) = 1$, $f(1) = 3$, $f(2) = 7$, $f(3) = 15$. Find $(f^{-1})^{-1}(2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q5/5

SET 78 • HSF-BF.B.4.c

FORWARD 2

Restrict the domain to make a function invertible • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

$f(x) = (x - 3)^2$ is not invertible on all real numbers. Which restriction and inverse are correct?

A. $x \geq 3;\ f^{-1}(x) = 3 + \sqrt{x}$

B. $x \geq 3;\ f^{-1}(x) = 3 - \sqrt{x}$

C. $x \geq 0;\ f^{-1}(x) = \sqrt{x} + 3$

D. $x \leq 3;\ f^{-1}(x) = 3 + \sqrt{x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q1/5

$h(x) = x^2$ has $h(-2) = 4$, $h(0) = 0$, and $h(2) = 4$. Why is $h^{-1}(4)$ not defined, and how can the domain be restricted to fix it?

A. 4 is the output of two inputs (−2 and 2); restrict h to $x \geq 0$ (or to $x \leq 0$).

B. 0 has no inverse, so remove $x = 0$ from the domain of h.

C. 4 is the output of two inputs (−2 and 2); restrict h to $-2 \leq x \leq 2$.

D. $h^{-1}(4) = 1/4$, so it is defined, and no restriction is needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q2/5

True or false?

Restricting the domain can turn a non-invertible function into an invertible one.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q3/5

$f(x) = x^2$ restricted to $x \leq 0$ has inverse

A. $\sqrt{x}$

B. x^2

C. $-\sqrt{x}$

D. $1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q4/5

$f(x) = (x - 3)^2$ has range $y \geq 0$. When its domain is restricted to make it invertible, why should the restricted function still take every output value $y \geq 0$?

A. So the restricted function has more outputs than the original function, which makes the inverse larger.

B. So the inverse is defined for every output of the original function; for example, $x \geq 3$ still gives every $y \geq 0$.

C. Because the inverse function must be identical to the original function on the restricted domain.

D. Because a restricted function must always have exactly the same domain as the original function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q5/5

SET 79 • HSF-BF.B.4.d • ADVANCED EXTENSION (+)

Restrict a domain to make a function invertible and write the inverse

MAIN	Algebra II main question	**HSF-BF.B.4.d**
BACKWARD 1	Find the vertex of a quadratic	**HSF-IF.C.8.a**
BACKWARD 2	Write inverses of one-to-one functions	**HSF-BF.B.4.a**
BACKWARD 3	Domains and ranges	**HSF-IF.B.5**
FORWARD 1	Compare different restrictions	**HSF-BF.B.4.d**
FORWARD 2	Restrict trigonometric functions to define inverses	**HSF-TF.B.6**

SET 79 • HSF-BF.B.4.d

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

Restrict $f(x) = (x - 3)^2$ to $x \geq 3$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = \sqrt{x} + 3,\ x \geq 3$

B. $f^{-1}(x) = \sqrt{(x + 3)},\ x \geq -3$

C. $f^{-1}(x) = \sqrt{x} - 3,\ x \geq 0$

D. $f^{-1}(x) = \sqrt{x} + 3,\ x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q1/5

Restrict $f(x) = x^2 + 1$ to $x \le 0$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = \sqrt{(x - 1)},\ x \ge 1$

B. $f^{-1}(x) = -\sqrt{(x + 1)},\ x \ge -1$

C. $f^{-1}(x) = -\sqrt{(x - 1)},\ x \ge 1$

D. $f^{-1}(x) = -\sqrt{(x - 1)},\ x \le 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q2/5

Restrict $f(x) = |x + 2|$ to $x \geq -2$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = x + 2,\ x \geq -2$

B. $f^{-1}(x) = x - 2,\ x \geq 0$

C. $f^{-1}(x) = -x - 2,\ x \geq 0$

D. $f^{-1}(x) = x - 2,\ x \geq -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q3/5

Restrict $f(x) = (x + 1)^2 - 4$ to $x \geq -1$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = -1 + \sqrt{(x + 4)}, x \geq -4$

B. $f^{-1}(x) = -1 - \sqrt{(x + 4)}, x \geq -4$

C. $f^{-1}(x) = -1 + \sqrt{(x + 4)}, x \geq -1$

D. $f^{-1}(x) = 1 + \sqrt{(x - 4)}, x \geq 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q4/5

Restrict $f(x) = x^4$ to $x \geq 0$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = x/4,\ x \geq 0$

B. $f^{-1}(x) = \pm\sqrt[4]{x},\ x \geq 0$

C. $f^{-1}(x) = \sqrt[4]{x},\ x \geq 0$

D. $f^{-1}(x) = 1/x^4,\ x > 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 1

Find the vertex of a quadratic • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a

Give the vertex of $y = (x + 1)^2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q1/5

Give the vertex of $y = x^2 + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q2/5

True or false?

A parabola is increasing on one side of its vertex and decreasing on the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q3/5

The vertex of $y = (x - 3)^2$ is

A. (0, 9)

B. (0, 3)

C. (−3, 0)

D. (3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q4/5

On which side of the vertex is $y = (x - 3)^2$ increasing?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 2

Write inverses of one-to-one functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

Solve $y = (x - 3)^2$ for x when $x \geq 3$.

A. $x = 3 - \sqrt{y}$

B. $x = \sqrt{(y + 3)}$

C. $x = \sqrt{y} - 3$

D. $x = 3 + \sqrt{y}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q1/5

Solve $y = x^2 + 1$ for x when $x \leq 0$.

A. $x = \sqrt{(y - 1)}$

B. $x = -\sqrt{y} - 1$

C. $x = -\sqrt{(y + 1)}$

D. $x = -\sqrt{(y - 1)}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q2/5

True or false?

When $x \leq 0$, the solution of $x^2 = 4$ is $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q3/5

Solving $y = x + 2$ $(x \geq -2)$ for x gives

A. $2 - y$

B. $y + 2$

C. $y - 2$

D. $-y - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q4/5

Solve $y = x^4$ for x when $x \geq 0$. Give x as a power of y, using ^ for exponents.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 3

Domains and ranges • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

Give the range of $f(x) = (x - 3)^2$ for $x \geq 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q1/5

Give the range of $f(x) = (x + 1)^2 - 4$ for $x \geq -1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q2/5

True or false?

The domain of the inverse is the range of the restricted function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q3/5

The range of $x^2 + 1$ for $x \leq 0$ is

A. $y \geq 0$

B. $y \geq 1$

C. all reals

D. $y \leq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q4/5

Give the range of $|x + 2|$ for $x \geq -2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q5/5

SET 79 • HSF-BF.B.4.d

FORWARD 1

Compare different restrictions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

Restrict $f(x) = (x - 3)^2$ to $x \leq 3$. Which is the inverse with its domain?

A. $f^{-1}(x) = 3 - \sqrt{x},\ x \leq 3$

B. $f^{-1}(x) = 3 + \sqrt{x},\ x \geq 0$

C. $f^{-1}(x) = 3 - \sqrt{x},\ x \geq 0$

D. $f^{-1}(x) = \sqrt{x} - 3,\ x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q1/5

A ball's height is $h(t) = -5(t - 2)^2 + 20$ for $0 \le t \le 4$. Which restriction and inverse give the time as a function of height on the way down?

A. $2 \le t \le 4;\ t = 2 + \sqrt{((20 - h)/5)}$

B. $2 \le t \le 4;\ t = 2 - \sqrt{((20 - h)/5)}$

C. $0 \le t \le 4;\ t = 2 + \sqrt{((20 - h)/5)}$

D. $0 \le t \le 2;\ t = 2 - \sqrt{((20 - h)/5)}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q2/5

True or false?

A function can have more than one valid restriction that makes it invertible.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q3/5

Which restriction does not make $f(x) = x^2$ invertible?

A. $x \geq 0$

B. $x \leq 0$

C. $-1 \leq x \leq 1$

D. $1 \leq x \leq 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q4/5

Why is the restriction $x \geq 0$ natural for $f(x) = x^2$ in the area formula $A = s^2$?

A. x^2 is always increasing.

B. Side lengths are nonnegative.

C. Squares have four sides.

D. Areas can be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q5/5

SET 79 • HSF-BF.B.4.d

FORWARD 2

Restrict trigonometric functions to define inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

Why is sin x not invertible on all real numbers?

A. It repeats values (for example sin 0 = sin π = 0), so it is not one-to-one.

B. Its outputs are larger than 1 for large x, outside the range of an inverse.

C. It is not defined for negative x, so an inverse would be incomplete.

D. It is an odd function, and odd functions never have an inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q1/5

What is the standard restricted domain of sin x, and why does it work?

A. $0 \le x \le \pi$: sin is increasing there and takes every value from -1 to 1 once.

B. $-\pi/2 \le x \le \pi/2$: sin is decreasing there and takes every value from 0 to 1 once.

C. $0 \le x \le 2\pi$: sin takes every value from -1 to 1 there, which is all that is needed.

D. $-\pi/2 \le x \le \pi/2$: sin is increasing there and takes every value from -1 to 1 once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q2/5

True or false?

cos x is restricted to $0 \leq x \leq \pi$ to define arccos.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q3/5

arcsin(1) equals

A. 0

B. 3π/2

C. π/2

D. π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q4/5

Why would $0 \le x \le \pi$ not work as a restricted domain for an inverse of sine?

A. sin is not defined at $x = \pi$, so the interval is not closed.

B. sin is negative everywhere on that interval, so it misses positive values.

C. sin increases then decreases there, so values repeat ($\sin(\pi/6) = \sin(5\pi/6)$).

D. sin is increasing on that whole interval, so it is one-to-one there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q5/5

SET 80 • HSF-BF.B.5 • ADVANCED EXTENSION (+)

Use the inverse relationship of exponents and logarithms to solve equations

MAIN	Algebra II main question	**HSF-BF.B.5**
BACKWARD 1	Write exact logarithmic solutions of exponential equations	**HSF-LE.A.4**
BACKWARD 2	Check for extraneous solutions	**HSA-REI.A.2**
BACKWARD 3	Domains of logarithmic expressions	**HSF-IF.B.5**
FORWARD 1	Use log properties and change of base in equations	**HSF-BF.B.5**
FORWARD 2	Check logarithmic solutions graphically	**HSA-REI.D.11**

SET 80 • HSF-BF.B.5

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

Solve $\log_3 x = 4$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q1/5

Solve $2\log_5(x - 1) = 4$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q2/5

Solve $\ln(x + 2) = 1$ exactly.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q3/5

Solve $\log_2 x + \log_2(x - 2) = 3$. The equation $x(x - 2) = 8$ gives two candidates. Give the actual solution and the extraneous value.

Solution: $x =$ _____

Extraneous value: $x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q4/5

Solve $\log x - \log(x - 9) = 1$ (base 10) and check.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 1

Write exact logarithmic solutions of exponential equations • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $3^x = 81$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q1/5

Solve $e^x = 5$ exactly.

A. $x = 5/e$

B. $x = \ln 5$

C. $x = e^5$

D. $x = \log 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q2/5

True or false?

$\log_b y = x$ means $b^x = y$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q3/5

$5^2 = 25$ written as a logarithm is

A. $\log_5 2 = 25$

B. $\log_2 25 = 5$

C. $\log_{25} 5 = 2$

D. $\log_5 25 = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q4/5

Solve $10^x = 1000$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 2

Check for extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Check $x = -2$ in $\log_2 x + \log_2(x - 2) = 3$.

A. $\log_2(-2) = 1$, so the left side is 3 and $x = -2$ is a solution.

B. $\log_2(-2) = -1$, so $x = -2$ is a solution.

C. $\log_2(-2)$ is undefined, so $x = -2$ is not a solution.

D. $(-2)(-4) = 8 = 2^3$, so $x = -2$ is a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q1/5

Check $x = 4$ in $\log_2 x + \log_2(x - 2) = 3$: find the value of the left side, then say whether $x = 4$ is a solution.

Left side: _____

Solution? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q2/5

True or false?

Combining logs can introduce solutions outside the original domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q3/5

Which value is excluded from $\log(x - 9)$?

A. $x = 20$

B. $x = 100$

C. $x = 5$

D. $x = 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q4/5

Solve $x(x - 2) = 8$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 3

Domains of logarithmic expressions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

Give the domain of $\log_5(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q1/5

Give the domain of ln(x + 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q2/5

True or false?

$\log 0 = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q3/5

The domain of $\log_2 x + \log_2(x - 2)$ is

A. $x > 2$

B. $x \neq 2$

C. $x > 0$

D. all reals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q4/5

Give the domain of $\log x - \log(x - 9)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q5/5

SET 80 • HSF-BF.B.5

FORWARD 1

Use log properties and change of base in equations • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

Solve $\log_2(x + 6) - \log_2 x = 2$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q1/5

Solve $\log_4 x = \log_2 3$. (Hint: change of base.)

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q2/5

True or false?

$\log_2(x + 6) - \log_2 x = \log_2((x + 6)/x)$ for $x > 0$.

A. True

B. False

$\log(x + 1) + \log(x - 1) = \log 8$ has solution

A. no solution

B. $x = 8$

C. $x = 3$

D. $x = -3$ and 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q4/5

Why does $\log_2(x - 5) = \log_2(1 - x)$ have no solution?

A. Both sides are defined only if $x > 5$ and $x < 1$ at the same time, which is impossible.

B. Setting $x - 5 = 1 - x$ gives $x = 3$, which is a solution, so the equation does have one.

C. The equation has a solution only when x is negative.

D. Logarithms with base 2 can never be equal to each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F1-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q5/5

SET 80 • HSF-BF.B.5

FORWARD 2

Check logarithmic solutions graphically • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

How do the graphs of $y = \log_2 x + \log_2(x - 2)$ and $y = 3$ show that the only solution is $x = 4$?

A. They intersect only at $x = 4$; the left graph exists only for $x > 2$, so there is no point at $x = -2$.

B. They intersect at $x = 4$ and at $x = -2$, so both values are solutions of the equation.

C. They never intersect, so $x = 4$ must come from the algebra only.

D. They intersect at $y = 4$, which is the solution of the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q1/5

Use a graph or table to approximate the solution of $\ln x = 2 - x$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q2/5

True or false?

An extraneous solution of a log equation corresponds to no intersection point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q3/5

The graphs of $y = \log x$ and $y = 1$ meet at

A. $x = e$

B. $x = 10$

C. $x = 1$

D. $x = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q4/5

Why can $\ln x = 2 - x$ not be solved exactly with logarithm rules?

A. $\ln x$ is undefined for every x, so the equation has no solution at all.

B. Logarithm rules apply only to equations with base-10 logarithms.

C. The equation has infinitely many solutions, so no single exact value exists.

D. x appears both inside a logarithm and outside it, so no algebraic step isolates x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q5/5

SET 81 • HSF-TF.A.3 • ADVANCED EXTENSION (+)

Derive values from special triangles and relate π − x, π + x, 2π − x to x

MAIN	Algebra II main question	**HSF-TF.A.3**
BACKWARD 1	Special right triangle side ratios	**HSG-SRT.C.6**
BACKWARD 2	Unit-circle values in all quadrants	**HSF-TF.A.2**
BACKWARD 3	The Pythagorean Theorem	**8.G.B.7**
FORWARD 1	Derive more special values and the reflection identities	**HSF-TF.A.3**
FORWARD 2	Explain symmetry and periodicity	**HSF-TF.A.4**

SET 81 • HSF-TF.A.3

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3

A 30°−60°−90° triangle with hypotenuse 1 is half of an equilateral triangle with side 1. Which derivation of sin(π/3) and cos(π/3) is correct?

A. The short leg is 1/2 and the long leg is $\sqrt{(1 - 1/4)} = \sqrt{3}/2$. The angle π/3 is opposite the long leg, so $\sin(\pi/3) = \sqrt{3}/2$ and $\cos(\pi/3) = 1/2$.

B. The long leg is $\sqrt{(1 + 1/4)} = \sqrt{5}/2$, so $\sin(\pi/3) = \sqrt{5}/2$.

C. The short leg is 1/2 and the long leg is $\sqrt{(1 - 1/4)} = \sqrt{3}/2$. The angle π/3 is opposite the short leg, so $\sin(\pi/3) = 1/2$ and $\cos(\pi/3) = \sqrt{3}/2$.

D. The legs are 1/2 and $1 - 1/2 = 1/2$, so $\sin(\pi/3) = \cos(\pi/3) = 1/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q1/5

Which derivation of $\tan(\pi/4)$ from a $45°-45°-90°$ triangle is correct?

A. The hypotenuse is twice a leg, so $\tan(\pi/4) = 1/2$.

B. $\tan(\pi/4) = \sin(\pi/4) + \cos(\pi/4) = \sqrt{2}$.

C. The legs are 1 and $\sqrt{2}$, so $\tan(\pi/4)$ = opposite/adjacent = $1/\sqrt{2} = \sqrt{2}/2$.

D. The two legs are equal, so $\tan(\pi/4)$ = opposite/adjacent = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q2/5

Which uses $\sin(\pi - x) = \sin x$ to find $\sin(5\pi/6)$ exactly, with a correct unit-circle explanation?

A. $\sin(5\pi/6) = -\sin(\pi/6) = -1/2$: the points for x and $\pi - x$ are reflections over the x-axis, so their y-coordinates are opposite.

B. $\sin(5\pi/6) = \sin(\pi/6) = 1/2$: the points for x and $\pi - x$ are reflections over the y-axis, so they have the same y-coordinate.

C. $\sin(5\pi/6) = \cos(\pi/6) = \sqrt{3}/2$: the point for $\pi - x$ is a quarter turn from the point for x, so y becomes x.

D. $\sin(5\pi/6) = \sin(\pi/6) = 1/2$: the points for x and $\pi - x$ are reflections over the x-axis, so they have the same y-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-M-Q3**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q3/5

Use $\cos(\pi + x) = -\cos x$ to find $\cos(7\pi/6)$ exactly.

$\cos(7\pi/6) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q4/5

Use $\tan(2\pi - x) = -\tan x$ to find $\tan(5\pi/3)$ exactly.

$\tan(5\pi/3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 1

Special right triangle side ratios • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

A 30°−60°−90° triangle has hypotenuse 2. Give the lengths of its legs exactly.

Short leg: _____

Long leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q1/5

Give the hypotenuse of a 45°−45°−90° triangle with legs 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q2/5

True or false?

In a 30°−60°−90° triangle, the shortest side is half the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q3/5

cos 30° equals

A. √3

B. √2/2

C. 1/2

D. √3/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q4/5

Give tan 60°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 2

Unit-circle values in all quadrants • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Give $\cos(5\pi/6)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q1/5

Give $\sin(7\pi/6)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q2/5

True or false?

sin is positive in quadrant II.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q3/5

Which quadrant contains 5π/3?

A. II

B. I

C. IV

D. III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q4/5

Give $\tan(\pi/3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 3

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

A right triangle has hypotenuse 1 and leg 1/2. Find the other leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q1/5

A right isosceles triangle has hypotenuse 1. Find each leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q2/5

True or false?

$(1/2)^2 + (\sqrt{3}/2)^2 = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q3/5

Legs 1 and 1 give hypotenuse

A. 2

B. $\sqrt{3}$

C. $\sqrt{2}$

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q4/5

Find x if $x^2 + x^2 = 1$ ($x > 0$).

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q5/5

SET 81 • HSF-TF.A.3

FORWARD 1

Derive more special values and the reflection identities • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3

Which derivation of $\tan(\pi/6)$ from a special triangle is correct?

A. The opposite side is half the hypotenuse, so $\tan(\pi/6) = 1/2$.

B. In the $30°-60°-90°$ triangle with hypotenuse 1, $\pi/6$ is opposite the leg $1/2$ and adjacent to the leg $\sqrt{3}/2$, so $\tan(\pi/6) = (1/2)/(\sqrt{3}/2) = 1/\sqrt{3} = \sqrt{3}/3$.

C. In the $30°-60°-90°$ triangle with hypotenuse 1, $\pi/6$ is opposite the leg $\sqrt{3}/2$ and adjacent to the leg $1/2$, so $\tan(\pi/6) = (\sqrt{3}/2)/(1/2) = \sqrt{3}$ exactly.

D. $\tan(\pi/6) = (1/2)/(\sqrt{3}/2) = \sqrt{3}/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q1/5

Which unit-circle argument explains why $\cos(\pi - x) = -\cos x$?

A. The points for x and $\pi - x$ are reflections of each other over the y-axis, so their x-coordinates (the cosines) are opposites.

B. $\cos(\pi - \pi/3) = -1/2 = -\cos(\pi/3)$, so the rule always holds.

C. $\cos(\pi - x) = \cos \pi - \cos x = -1 - \cos x$.

D. The points for x and $\pi - x$ are reflections of each other over the x-axis, so their x-coordinates (the cosines) are opposites.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q2/5

True or false?

$\sin(\pi + x) = -\sin x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q3/5

$\cos(2\pi - x)$ equals

A. $\cos x$

B. $-\cos x$

C. $\sin x$

D. $-\sin x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q4/5

Use $\sin(2\pi - x) = -\sin x$ to find $\sin(11\pi/6)$ exactly.

$\sin(11\pi/6) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSF-TF.A.3

FORWARD 2

Explain symmetry and periodicity • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Which unit-circle reasoning explains why $\sin(-x) = -\sin x$?

A. The points for x and −x are opposite points, so their x-coordinates are equal.

B. The points for x and −x are reflections over the y-axis, so their y-coordinates are opposites.

C. The points for x and −x are reflections over the x-axis, so their y-coordinates are opposites.

D. The points for x and −x are the same point, so their y-coordinates are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q1/5

Is cos an even or odd function, and why?

A. Even: the points for x and −x have the same y-coordinate, so cos(−x) = cos x.

B. Even: the points for x and −x have the same x-coordinate, so cos(−x) = cos x.

C. Odd: the points for x and −x have opposite x-coordinates, so cos(−x) = −cos x.

D. Neither: cos(−x) is not related to cos x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q2/5

True or false?

$\sin(x + 2\pi) = \sin x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q3/5

$\tan(-\pi/4)$ equals

A. undefined

B. -1

C. 1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q4/5

Using $2\pi - x = -x + 2\pi$, which reasoning explains why $\tan(2\pi - x) = -\tan x$?

A. $\tan(-x + 2\pi) = \tan(-x)$ by periodicity, and tan is odd, so $\tan(-x) = -\tan x$.

B. $\tan(-x + 2\pi) = \tan x$ by periodicity, and tan is even.

C. $\tan(-x + 2\pi) = \tan(-x) + \tan(2\pi) = -\tan x + 1$.

D. $\tan(2\pi - x) = 2\pi - \tan x$, because tan distributes over subtraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q5/5

SET 82 • HSF-TF.A.4 • ADVANCED EXTENSION (+)

Explain the symmetry and periodicity of trigonometric functions with the unit circle

MAIN	Algebra II main question	**HSF-TF.A.4**
BACKWARD 1	Explain the unit-circle definition	**HSF-TF.A.2**
BACKWARD 2	Even and odd functions	**HSF-BF.B.3**
BACKWARD 3	Reflect points over the axes	**8.G.A.3**
FORWARD 1	Use periodicity and symmetry to evaluate and simplify	**HSF-TF.A.4**
FORWARD 2	Restrict trigonometric functions so they have inverses	**HSF-TF.B.6**

SET 82 • HSF-TF.A.4

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Which unit-circle explanation shows why $\sin(-x) = -\sin x$?

A. The points for x and $-x$ are (a, b) and $(-a, b)$, reflections of each other over the y-axis, so their sines are opposites.

B. $\sin(-\pi/6) = -1/2 = -\sin(\pi/6)$, so the rule always holds.

C. The points for x and $-x$ are (a, b) and $(a, -b)$, reflections over the x-axis, so the sines b and $-b$ are opposites.

D. A negative sign can always be moved outside a function, so $\sin(-x) = -\sin x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q1/5

Which unit-circle explanation shows why $\cos(x + 2\pi) = \cos x$?

A. Adding 2π moves the point to the opposite side of the unit circle, and opposite points on the circle have the same x-coordinate, so the cosine is the same.

B. $\cos(x + 2\pi) = \cos x + \cos 2\pi = \cos x + 1$.

C. Adding 2π travels one full turn around the unit circle and ends at the same point, so the x-coordinate (the cosine) is the same.

D. $\cos(0 + 2\pi) = 1 = \cos 0$, so the rule always holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-M-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q2/5

Which unit-circle explanation shows why $\tan(x + \pi) = \tan x$?

A. Adding π gives the point $(a, -b)$, a reflection over the x-axis, so $\tan(x + \pi) = -b/a = -\tan x$, which is not $\tan x$.

B. Adding π gives the opposite point $(-a, -b)$ of (a, b), and $\tan = (-b)/(-a) = b/a$, the same ratio.

C. Adding π travels a full turn, so the point is the same.

D. $\tan(\pi/4 + \pi) = 1 = \tan(\pi/4)$, so the rule always holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q3/5

Use symmetry to evaluate $\cos(-2\pi/3)$.

A. $1/2$, because cos is even: $\cos(-2\pi/3) = \cos(\pi/3) = 1/2$

B. $-1/2$, because cos is even: $\cos(-2\pi/3) = \cos(2\pi/3) = -1/2$

C. $1/2$, because cos is odd: $\cos(-2\pi/3) = -\cos(2\pi/3) = 1/2$

D. $-\sqrt{3}/2$, because $\cos(-2\pi/3) = -\sin(2\pi/3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-M-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q4/5

Is $f(x) = \sin x \cos x$ even, odd, or neither?

A. Even: a product of two functions is always even

B. Odd: $f(-x) = \sin(-x)\cos(-x) = (-\sin x)(\cos x) = -f(x)$

C. Even: $f(-x) = \sin(-x)\cos(-x) = \sin x \cos x = f(x)$

D. Neither: sin is odd but cos is even

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q5/5

SET 82 • HSF-TF.A.4

BACKWARD 1

Explain the unit-circle definition • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Which point on the unit circle represents angle x?

A. (cos x, x)

B. (sin x, cos x)

C. (x, sin x)

D. (cos x, sin x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q1/5

In which direction from (1, 0) does a negative angle go on the unit circle? Answer Clockwise or Counterclockwise.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q2/5

True or false?

$\tan x = \sin x / \cos x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q3/5

The point for angle π is

A. $(0, -1)$

B. $(-1, 0)$

C. $(0, 1)$

D. $(1, 0)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q4/5

Give $\cos(2\pi/3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSF-TF.A.4

BACKWARD 2

Even and odd functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

Which is the definition of an even function?

A. $f(x + 2\pi) = f(x)$ for every x in the domain

B. $f(-x) = f(x)$ for every x in the domain

C. $f(-x) = -f(x)$ for every x in the domain

D. $f(x) > 0$ for every x in the domain

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q1/5

Which is the definition of an odd function?

A. $f(-x) = f(x)$ for every x in the domain

B. $f(x) < 0$ for every x in the domain

C. $f(x + \pi) = -f(x)$ for every x in the domain

D. $f(-x) = -f(x)$ for every x in the domain

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q2/5

True or false?

An odd function's graph is symmetric about the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q3/5

x^3 is

A. odd

B. neither

C. even

D. both

Is the product of an odd function and an even function odd or even? Answer Odd or Even.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q5/5

SET 82 • HSF-TF.A.4

BACKWARD 3

Reflect points over the axes • Grade 8 • Geometry • 8.G.A.3

Reflect (a, b) over the x-axis.

A. (b, a)

B. (−a, b)

C. (a, −b)

D. (−a, −b)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q1/5

Rotate (a, b) by 180° about the origin.

A. (−a, −b)

B. (a, −b)

C. (−a, b)

D. (−b, a)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q2/5

True or false?

Reflecting over the x-axis keeps the y-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q3/5

Reflecting (0.6, 0.8) over the x-axis gives

A. (0.8, 0.6)

B. (−0.6, 0.8)

C. (−0.6, −0.8)

D. (0.6, −0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q4/5

Rotate the point $(\sqrt{3}/2, 1/2)$ by 180° about the origin.

A. $(\sqrt{3}/2, -1/2)$

B. $(-\sqrt{3}/2, -1/2)$

C. $(-1/2, \sqrt{3}/2)$

D. $(-\sqrt{3}/2, 1/2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q5/5

SET 82 • HSF-TF.A.4

FORWARD 1

Use periodicity and symmetry to evaluate and simplify • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Find $\sin(17\pi/6)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q1/5

Find $\tan(-11\pi/4)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q2/5

True or false?

$\cos(x + \pi) = -\cos x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q3/5

$\sin(x + \pi) + \sin x$ equals

A. 1

B. $\sin 2x$

C. $2 \sin x$

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q4/5

Simplify $\cos(-x) + \cos(x + 2\pi)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q5/5

SET 82 • HSF-TF.A.4

FORWARD 2

Restrict trigonometric functions so they have inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

How does periodicity prevent sin x from having an inverse on all real numbers?

A. Each value appears only once in every period, so periodicity changes nothing.

B. Periodicity makes sin increasing everywhere, so it has no turning points.

C. Each value repeats every 2π, so an output does not determine its input.

D. Periodic functions are undefined at multiples of 2π, which breaks the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q1/5

What is the restricted domain used for tan x, and why?

A. $-\pi/2 < x < \pi/2$: tan increases there and takes every real value once.

B. $0 < x < \pi$: tan is defined and increasing everywhere there.

C. $-\pi/2 \leq x \leq \pi/2$: tan is defined at both endpoints.

D. $0 \leq x < 2\pi$: tan takes every real value there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q2/5

True or false?

arccos gives values between 0 and π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q3/5

arctan(−1) equals

A. 7π/4

B. 3π/4

C. −π/4

D. π/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q4/5

Why is $\arcsin(-1/2) = -\arcsin(1/2)$?

A. $\arcsin(-1/2) = 7\pi/6$ because $\sin(7\pi/6) = -1/2$, which is $-\arcsin(1/2)$.

B. sin is even on $[-\pi/2, \pi/2]$, so $\sin(-\pi/6) = 1/2$, giving $\arcsin(-1/2) = \pi/6$.

C. A minus sign can always be moved outside any function, so $\arcsin(-a) = -\arcsin(a)$.

D. sin is odd on $[-\pi/2, \pi/2]$, so $\sin(-\pi/6) = -1/2$, giving $\arcsin(-1/2) = -\pi/6 = -\arcsin(1/2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q5/5

SET 83 • HSF-TF.B.6 • ADVANCED EXTENSION (+)

Choose and justify domain restrictions that make trigonometric functions invertible

MAIN	Algebra II main question	**HSF-TF.B.6**
BACKWARD 1	Restrict a domain to make a function invertible	**HSF-BF.B.4.d**
BACKWARD 2	Exact values on the unit circle	**HSF-TF.A.2**
BACKWARD 3	Periodicity of trigonometric functions	**HSF-TF.A.4**
FORWARD 1	Compose trigonometric functions with their inverses	**HSF-TF.B.6**
FORWARD 2	Solve trigonometric equations with inverses	**HSF-TF.B.7**

SET 83 • HSF-TF.B.6

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

Why does sin x have no inverse on all real numbers, and on which restricted domain does it have one?

A. sin repeats values, so it is not one-to-one; on $0 \le x \le \pi$ it is increasing and takes each value in $[-1, 1]$ once.

B. sin is not defined for all real numbers; on $0 \le x \le \pi$ it is defined and takes each value in $[-1, 1]$ once.

C. sin repeats values, so it is not one-to-one; on $-\pi/2 \le x \le \pi/2$ it is increasing and takes each value in $[-1, 1]$ once.

D. sin has outputs greater than 1 for large x; on $-\pi \le x \le \pi$ it takes each value in $[-1, 1]$ exactly once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q1/5

What is the standard restricted domain of cos x, and why does it work?

A. $0 \le x \le 2\pi$: cos takes every value from -1 to 1 there, so it is enough.

B. $0 \le x \le \pi$: cos decreases from 1 to -1 there, taking each value once.

C. $-\pi/2 \le x \le \pi/2$: cos increases from -1 to 1 there, taking each value once.

D. $-\pi \le x \le 0$: cos decreases from 1 to -1 there, taking each value once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q2/5

What is the standard restricted domain of tan x, and why does it work?

A. $-\pi/2 < x < \pi/2$: tan increases from $-\infty$ to ∞ there, taking each real value once.

B. $0 < x < \pi$: tan is defined and increasing at every point of that interval.

C. $0 \le x < \pi/2$: tan increases there and takes each real value exactly once.

D. $-\pi/2 \le x \le \pi/2$: tan increases there, including its values at both endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q3/5

What is arcsin(1/2), and why is it not 5π/6?

A. π/6 and 5π/6: arcsin gives every angle in [0, 2π) whose sine is 1/2.

B. π/3: arcsin(1/2) is the angle whose cosine is 1/2, not 5π/6.

C. 5π/6: arcsin gives the largest angle in [0, 2π) whose sine is 1/2.

D. π/6: arcsin outputs only angles in [−π/2, π/2], and 5π/6 is outside that interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q4/5

Evaluate $\arccos(-1/2)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 1

Restrict a domain to make a function invertible • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

On which domain is $f(x) = x^2$ invertible?

A. all real numbers

B. $x \geq -2$

C. $x \geq 0$

D. $-1 \leq x \leq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q1/5

Why must a function be strictly increasing or strictly decreasing on an interval to be invertible there?

A. Then the function is defined at every real number, inside and outside the interval.

B. Then the function has no zeros on the interval, so it never crosses the x-axis.

C. Then the outputs are all positive on the interval, so each can be undone.

D. Then no output repeats on the interval, so each output comes from exactly one input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q2/5

True or false?

A function that is always increasing on an interval is one-to-one there.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q3/5

$f(x) = (x - 1)^2$ is one-to-one on

A. $-1 \leq x \leq 3$

B. all reals

C. $x \geq 1$

D. $x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q4/5

Which is the inverse of $f(x) = x^2$ on $x \geq 0$?

A. $-\sqrt{x}$

B. $\sqrt{x}$

C. $x/2$

D. $1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 2

Exact values on the unit circle • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Give $\sin(\pi/6)$ and $\sin(5\pi/6)$ exactly.

$\sin(\pi/6) =$ _____

$\sin(5\pi/6) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q1/5

Give $\cos(2\pi/3)$ and $\cos(4\pi/3)$ exactly.

$\cos(2\pi/3) =$ _____

$\cos(4\pi/3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q2/5

True or false?

$\tan(\pi/4) = \tan(3\pi/4)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q3/5

Which angle in [0, π] has cos = −1/2?

A. 2π/3

B. π/3

C. 4π/3

D. 5π/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q4/5

Give $\sin(-\pi/6)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 3

Periodicity of trigonometric functions • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

What is the period of sin x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q1/5

What is the period of tan x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q2/5

True or false?

A periodic function cannot be one-to-one on its whole domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q3/5

$\sin x = 0$ for

A. $x = \pi/2 + k\pi$

B. no x

C. $x = 2k\pi$ only

D. $x = k\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q4/5

Which two angles both have cosine 1?

A. 0 and 2π

B. 0 and π

C. π and 2π

D. π/2 and 3π/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q5/5

SET 83 • HSF-TF.B.6

FORWARD 1

Compose trigonometric functions with their inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

Evaluate $\arcsin(\sin(5\pi/6))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q1/5

Evaluate sin(arcsin(0.3)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q2/5

True or false?

arccos(cos x) = x for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q3/5

arctan(tan(3π/4)) equals

A. $-\pi/4$

B. $-3\pi/4$

C. $3\pi/4$

D. $\pi/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q4/5

What are the domain and range of arccos?

A. all real numbers; range $[0, \pi]$

B. domain $[0, \pi]$; range $[-1, 1]$

C. domain $[-1, 1]$; range $[-\pi/2, \pi/2]$

D. domain $[-1, 1]$; range $[0, \pi]$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSF-TF.B.6

FORWARD 2

Solve trigonometric equations with inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

Solve $\sin x = 0.4$ on $0 \leq x < 2\pi$. Give both solutions to the nearest hundredth, separated by a comma. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q1/5

Solve $\cos x = 0.4$ on $0 \leq x < 2\pi$. Give both solutions to the nearest hundredth, separated by a comma. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q2/5

True or false?

For $0 < c < 1$, arcsin c gives only one of the two solutions of $\sin x = c$ on $[0, 2\pi)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q3/5

The second solution of $\sin x = c$ in $[0, 2\pi)$ (with $0 < c < 1$) is

A. $\pi + \arcsin c$

B. $-\arcsin c$

C. $\pi - \arcsin c$

D. $2\pi - \arcsin c$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q4/5

For $-1 < c < 1$, how is the restricted range of arccos used to find both solutions of $\cos x = c$ on $[0, 2\pi)$?

A. arccos c gives the solution in $(0, \pi)$; the other is $2\pi - \arccos c$, by symmetry over the x-axis.

B. arccos c gives both solutions at once, because its range is the whole interval $[0, 2\pi)$.

C. arccos c gives the solution in $(0, \pi)$; the other is $\pi - \arccos c$, by symmetry over the y-axis.

D. arccos c gives the solution in $(-\pi/2, \pi/2)$; the other is $\arccos c + \pi$, by periodicity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q5/5

SET 84 • HSF-TF.B.7 • ADVANCED EXTENSION (+)

Solve trigonometric equations in models with inverse functions

MAIN	Algebra II main question	**HSF-TF.B.7**
BACKWARD 1	Evaluate inverse trigonometric functions on restricted domains	**HSF-TF.B.6**
BACKWARD 2	Parameters of sinusoidal models	**HSF-TF.B.5**
BACKWARD 3	Symmetry of unit-circle solutions	**HSF-TF.A.2**
FORWARD 1	Find all real solutions and solve equations that factor	**HSF-TF.B.7**
FORWARD 2	Solve equations with sum and double-angle identities	**HSF-TF.C.9**

SET 84 • HSF-TF.B.7

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

Solve $2\sin x - 1 = 0$ for $0 \leq x < 2\pi$. Give exact values.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q1/5

Solve $3\cos x = -2$ for $0 \leq x < 2\pi$. Give both solutions to the nearest hundredth, separated by a comma. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q2/5

A Ferris wheel rider's height is $h(t) = 12 + 10\sin(\pi t/15)$ m after t seconds. Find every time in $0 \le t < 30$ when h = 17 m, in seconds, separated by commas.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q3/5

Harbor depth is $d(t) = 5 + 4\cos(\pi t/6)$ ft after t hours. Find every time in $0 \le t < 12$ when $d = 7$ ft, in hours, separated by commas.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q4/5

Solve $\tan x = 3$ for $0 \leq x < 2\pi$. Give both solutions to the nearest hundredth, separated by a comma. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 1

Evaluate inverse trigonometric functions on restricted domains • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

Evaluate arcsin(1/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q1/5

Evaluate arccos(1/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q2/5

True or false?

arctan 3 is between 0 and π/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q3/5

arccos(−2/3) is in quadrant

A. II

B. III

C. I

D. IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q4/5

Evaluate arctan 3 to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 2

Parameters of sinusoidal models • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

In $h(t) = 12 + 10\sin(\pi t/15)$, what is the period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q1/5

A rider's height is $h(t) = 12 + 10\sin(\pi t/15)$ m. Setting $h(t) = 17$ gives an equation $\sin(\pi t/15) = c$. Find c.

$c =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q2/5

True or false?

$d(t) = 5 + 4\cos(\pi t/6)$ has a maximum of 5.

A. True

B. False

Isolating the cosine in $5 + 4\cos(\pi t/6) = 7$ gives

A. $\cos(\pi t/6) = 1/2$

B. $\cos(\pi t/6) = 3$

C. $\cos(\pi t/6) = 2$

D. $\cos(\pi t/6) = 7/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q4/5

What is the midline of d(t)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 3

Symmetry of unit-circle solutions • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Give both angles in $[0, 2\pi)$ with $\sin = 1/2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q1/5

Give both angles in [0, 2π) with cos = 1/2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q2/5

True or false?

$\tan x = c$ has two solutions in $[0, 2\pi)$ that differ by π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q3/5

For $-1 < c < 1$, $\cos x = c$ has a solution a in $(0, \pi)$. The other solution in $[0, 2\pi)$ is

A. $2\pi - a$

B. $\pi + a$

C. $-a$

D. $\pi - a$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q4/5

Give both angles in $[0, 2\pi)$ with $\tan = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q5/5

SET 84 • HSF-TF.B.7

FORWARD 1

Find all real solutions and solve equations that factor • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

Which gives all real solutions of $2\sin x - 1 = 0$?

A. $x = \pi/3 + 2k\pi$ or $x = 2\pi/3 + 2k\pi$, k an integer

B. $x = \pi/6 + k\pi$, k an integer

C. $x = \pi/6 + 2k\pi$ or $x = 5\pi/6 + 2k\pi$, k an integer

D. $x = \pi/6 + 2k\pi$, k an integer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q1/5

Solve $2\sin^2 x - \sin x = 0$ for $0 \leq x < 2\pi$. List all solutions exactly, separated by commas.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q2/5

True or false?

$\cos x = 2$ has no real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q3/5

How many solutions does $\sin(2x) = 1/2$ have in $0 \leq x < 2\pi$?

A. 2

B. 4

C. 8

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F1-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q4/5

Solve $\cos^2 x = 1/4$ for $0 \leq x < 2\pi$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q5/5

SET 84 • HSF-TF.B.7

FORWARD 2

Solve equations with sum and double-angle identities • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9

Use $\sin 2x = 2\sin x \cos x$ to solve $\sin 2x = \cos x$ for $0 \leq x < 2\pi$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q1/5

Use $\cos 2x = 1 - 2\sin^2 x$ to solve $\cos 2x = \sin x$ for $0 \leq x < 2\pi$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q2/5

True or false?

$\sin(x + \pi/2) = \cos x$.

A. True

B. False

cos 2x equals

A. $1 + 2\sin^2 x$

B. $\cos^2 x - \sin^2 x$

C. $\cos^2 x + \sin^2 x$

D. $2\cos x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q4**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q4/5

Why are identities useful for solving equations that contain different trigonometric functions?

A. They rewrite the equation in one function (or let it factor), so inverse functions can be applied.

B. They turn every equation into a linear equation that has exactly one solution.

C. They remove the need to check the solutions in the original equation.

D. They change the period of each function to 2π, so the graphs match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q5/5